AOS-CX 10.07 Monitoring Guide

6300, 6400 Switch Series

a Hewlett Packard Enterprise company

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Contents

Chapter 1
About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A)
- Aruba 6400 Switch Series (JL741A, R0X26A, R0X27A, R0X29A, R0X30A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in Support and other resources.

Command syntax notation conventions

Convention	Usage
`example-text`	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (`[ ]`).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ *<example-text>* ■ <example-text> ■ *example-text* ■ *example-text*	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
|	Vertical bar. A logical `OR` that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
`{ }`	Braces. Indicates that at least one of the enclosed items is required.

Convention	Usage
`[ ]`	Brackets. Indicates that the enclosed item or items are optional.
… or `...`	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

`switch>`

Indicates the operator command context.

`switch#`

Indicates the manager command context.

`switch(`*`CONTEXT-NAME`*`)#`

Indicates the configuration context for a feature. For example:

`switch(config-if)#`

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

`switch(config-vlan-`*`<VLAN-ID>`*`)#`

Where *`<VLAN-ID>`* is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

`member/slot/port`

On the 6300 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 10. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.

- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 on member 1.

On the 6400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
    - Management modules are on the front of the switch in slots 1/1 and 1/2.
    - Line modules are on the front of the switch starting in slot 1/3.
- *port*: Physical number of a port on a line module.

For example, the logical interface `1/3/4` in software is associated with physical port 4 in slot 3 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
    - *member*: 1.
    - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
    - *member*: 1.
    - *tray*: 1 to 4.
    - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:
    - *member*: 1.
    - *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Chapter 2
Monitoring hardware through visual observation

Confirming normal operation of the switch by reading LEDs

This task describes using the switch LEDs to confirm that the switch is operating normally.

Procedure

1. Quick check: Verify that the chassis has power and there are no fault conditions.

   On the front of the switch, verify that the states of the following LEDs are On Green:

   - Power
   - Health
2. Verify that the Health LEDs of all installed line modules are On Green.
3. Verify that the Health LEDs of all installed management modules are On Green.
4. Verify that the network ports are operating normally.
   a. On the active management module, check the Status Front section. Verify that each LED that indicates a line module is in one of the following states:
      - On Green (normal operation)
      - Off (no line module installed)

   b. On each line module, verify that each port LED is in one of the following states:
      - On Green, Half-Bright Green, or Flickering Green (normal operation)
      - Off (no cable connected or port off by default in config)
5. Verify that the power supplies are operating normally.
   a. On the active management module, check the Status Front section. Verify that each LED that indicates a power supply is in one of the following states:
      - On Green (normal operation)
      - Off (no power supply installed)

   b. On each power supply, verify that LEDs are in the following states:
      - Power LED: On Green
      - Fault LED: Off
6. Verify that the rear components are operating normally by checking the Status Rear section of the active management module:
   a. Verify that the LEDs for the fabric modules are in one of the following states:
      - On Green (normal operation)
      - Off (component not installed)

   b. Verify that the LEDs for the fan trays and fans are On Green.
7. Verify that the standby management module is ready to take over as the active management module. On the standby management module, verify the states of the following LEDs:

- Health LED is On Green.
- Management state standby (Stby) LED is On Green.

Detecting if the switch is not ready for a failover event

This task describes using the switch LEDs to detect if the switch is not ready for the loss of a fabric module or for a failover from the active management module to the standby management module.

Although you can detect power supply failures by viewing the LEDs, you must use software commands to determine if the power supply redundancy is sufficient to power the chassis if a power supply fails.

Procedure

1. Detect if the standby management module is shut down.

   If the standby management module is shut down, the LED states are as follows:
   - The standby management module health LED is Off.
   - The standby management state active (Actv) LED is Off.
   - The standby management state standby (Stby) LED is Off.
   - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Off. For example, if the active management module is Management Module LED 5, Management Modules LED 6 is Off.
2. Detect if the standby management module is in a transient state. If the standby management module is booting, updating, or in another transient state, the LED states are as follows:
   - The standby management module health LED is Slow Flash Green when the service operating system is running or during an operating system update.
   - The standby management module Booting LED is Slow Flash Green when the ArubaOS-CX operating system is booting.
   - The standby management state active (Actv) LED is Off.
   - The standby management state standby (Stby) LED is Off.
   - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Slow Flash Green.
3. Detect if a fabric module is shut down or not present. If a fabric module is shut down or not present, the LED states are as follows:
   - On the active management module, in the Status Rear section, the LED for the fabric module is Off.
   - On the rear display module, the LED for the fabric module is Off.
   - On the fabric module, the health LED is Off. However, the fabric module is behind fan 1 and is not directly visible.

Finding faulted components using the switch LEDs

This task describes using the switch LEDs to find components that are in a fault condition.

All green LEDs—except for chassis power LEDs and the Usr1 LED—are off when the LED mode is set to Light Faults (The Usr1 LED of the LED Mode section of the active management module is On Green and the default behavior for the Usr1 LED is being used.).

Procedure

1. Find the switch that has the fault condition, which is indicated by a chassis health LED in the state of Slow Flash Orange.

   The chassis health LED is located on the front of the switch and on the rear panel of the switch.
2. If you are at the back of the switch, on the rear panel, look for LEDs that are in the Slow Flash Orange state:

   The Status Rear area has LEDs for power supplies, fabric modules, fan trays, and fans. The number on the LED represents the unit number of the component.

   If the only LED in a state of Slow Flash Orange is the Chassis health LED, go to the front of the switch.
3. At the front of the switch, on the active management module, look for LEDs that are in the Slow Flash Orange state:
   - The Status Front area has LEDs for power supplies, line and fabric modules, and management modules. The number on the LED indicates the slot number of the component.
   - The Status Rear area has LEDs for fabric modules and fan trays, with a single LED for all the fans in the fan tray. The number on the LED represents the slot or bay number of the component.
4. Use the number indicated by the LED that is flashing to locate the slot that contains the faulted component.

   The fabric modules are located behind the fan trays, and the fabric module number corresponds to the fan tray number.
5. At the front of the switch, on line modules, look for LEDs that are in the Slow Flash Orange state:

   Module LEDs and Port LEDs indicate faults if their states are Slow Flash Orange.

Chapter 3
Aruba 6300/6400 Switch Series LEDs

Switch and port LEDs for 6300 Switch Series

Figure 1 *Switch and Port LEDs*

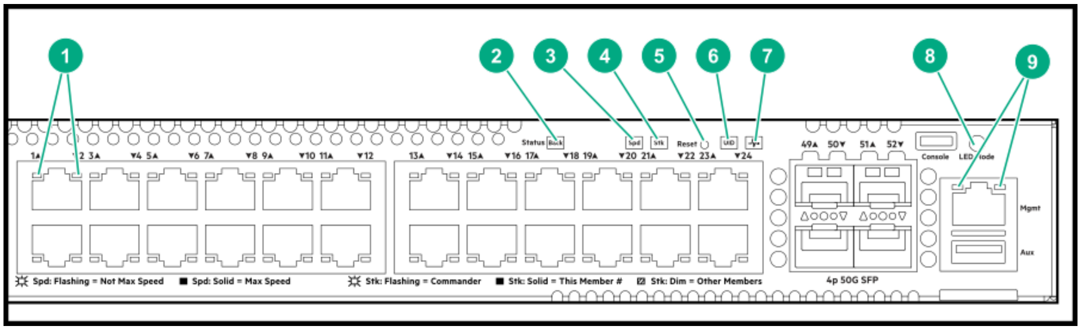


Table 1: *Switch and port LEDs: Labels and description*

Label	Description
1	Switch port LEDs
2	Back Module status LED
3	Speed mode selected LED
4	PoE mode selected
5	Reset button Usr mode selected LED
6	UID (Unit Identification)
7	Global Status LED
8	LED Mode status LED
9	Management Console LED

Table 2: *Front panel LED behavior*

Switch LEDs	Function	State	Meaning
Back LED	Status of modular components installed in the back of the chassis (not applicable for 6300F switches)	On - Green	Normal
		Slow Flash - Amber	Fault in one of the modules in the back of the chassis

Switch LEDs	Function	State	Meaning
PoE LED	Indicates Port LEDs are showing PoE information (not applicable for non PoE switches)	Off	PoE mode not selected
		On - Green	PoE mode selected
		Slow Flash - Amber	Hardware failure PoE enabled port, PoE mode not selected
		On - Amber	Hardware failure PoE enabled port, PoE mode selected
Spd LED	Indicates Port LEDs are showing speed information	Off	Speed mode not selected
		On - Green	Speed mode selected
		Not Implemented	No fault defined
Stk LED	Indicates Port LEDs are showing stacking mode information	Off	Stacking mode not selected
		On - Green	Stacking mode selected
		On - Amber	A port has a stacking failure. Stacking mode selected
		Slow flash Amber	A port has a stacking failure. Stacking mode not selected
UID LED	User-configurable LED	Off	User defined the located LED : OFF
		On/Flash Blue (for 30 min)	User defined the locator LED: On/Flash
Global Status Indicator LED	Overall status of the product	Flash - Green	Self-test in progress during UBOOT, SVOS and ArubaOS-CV
		On - Green	Successfully initialized ArubaOS-CX
		Flash - Amber	Recoverable faults (e.g. fans, PSU fault)
		On - Amber	Critical faults (e.g. exceed temperature limit)

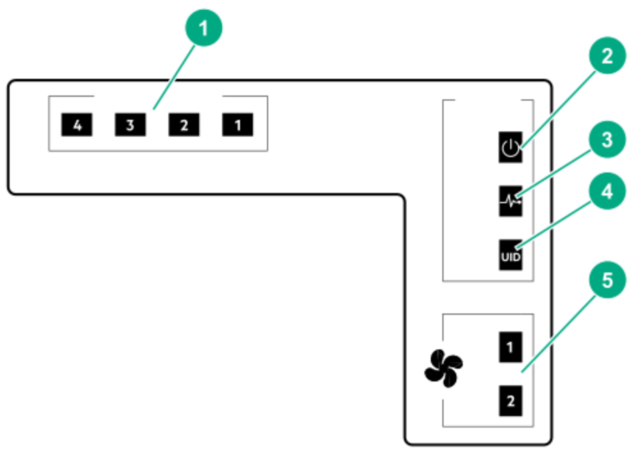
Switch LEDs	Function	State	Meaning
OOBM Status Indicator LED	Status of OOBM Link connectivity	Off	OOBM port is not connected, no link established
		Half Bright - Green	OOBM port is enabled and established link with partner
		On - Green	Experiencing high bandwidth utilization
		Activity Flicker - Green	% of the time that the LED light up is roughly proportional to the % of full bandwidth utilization of the port
* Press the Mode Select button to switch between User(default), PoE, Spd, or Stk Mode.			

Table 3: *Rear Panel LED behavior*

Switch LEDs	Function	State/Mode	Meaning
Fan health LED	Status of fan	On - Green	Normal
		Slow flash - Amber	Fan fault
UID LED	User-configurable LED	Off	User define the locator LED : OFF
		On/Flash (30 min) - blue	User define the locator LED: On/Flash
PSU Status Indicator LED	Status of power supply	On Green	Normal
		Off	No power, PSU has invalid AC input of invalid DC outputs
		Slow Flash - Green	Power supply has faulted or warning

Switch and port LEDs for 6400 Switch Series

Figure 2 *Rear panel LEDs for 6400 Switch Series*

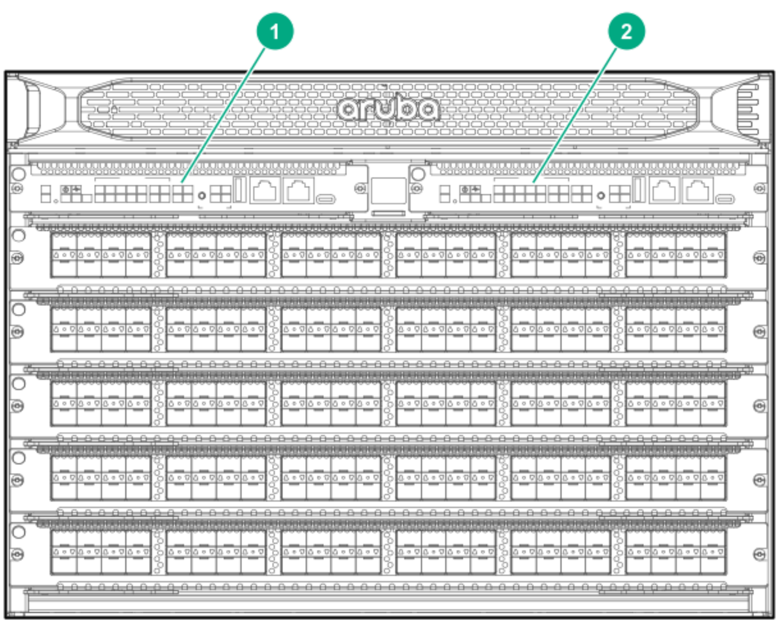


1	Power supply status (1) (2) (3) (4)
2	Chassis power LED
3	Chassis health LED
4	Unit identification (UID) LED
5	Fan tray status (1, 2)

Front panel LEDs for 6400 Switch Series

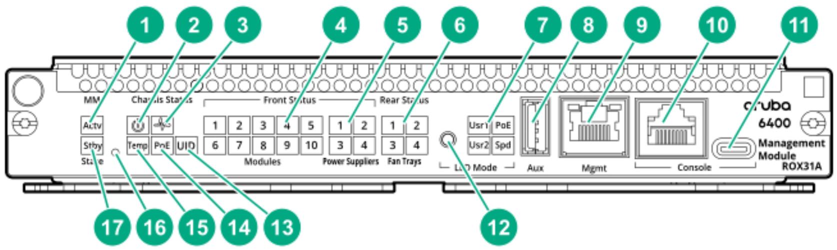
The Aruba 6400 switches have two management module (MM) slots. Management modules support control plane activities and in-memory running of the Time Series Database.

Figure 3 *Management module slots with management modules installed*



When two management modules are installed, one operates in active mode and the other operates in standby mode. The active slot is determined by election. Installing two management modules provides control plane high availability.

Figure 4 *Management module features*



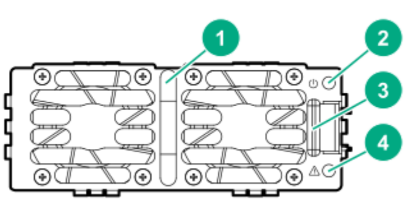
1	Mgmt state (Actv) LED	Indicates the status of the management module after booting. If the MM is the active MM, then the LED glows steady green.
2	System power LED	When the system is receiving power, glows steady green.=.
3	Management module health LED (green)	Indicates status of the switch. LED glows steady green when switch is ready after booting from the Network Operating System (NOS).
4	Line module status LEDs	Indicates if a line module is installed in a line module slot (1 through 5 for 6405 switches; 1 through 10 on 6410 switches). If a line module is installed in a given slot, then the numbered LED for that slot glows steady green.
5	Front Power supply status (1 2 3 4) LEDs	Indicates if a power supply is installed in the slot. If an active power supply is installed, then the LEDs glow steady green.
6	Fan tray status LEDs (1 - 4)	Indicate if the fan tray is installed in the slot. If a fan tray is installed in the slot, then the LED glows steady green.
7	LED mode: Usr1, Usr2 Spd, and PoE LEDs	The display of these LEDs is based on the LED mode button selection. ■ Usr1 LED: Indicates if the line module is working correctly. ■ Usr2 LED: Reserved ■ Spd LED: Indicates the traffic rate of the line module.
8	Auxillary port	Without a USB device installed, the auxiliary port LED is off after power-on and self-test. With a USB device installed, this LED displays the following after power-on and self-test: ■ Steady green: USB installed, initialized, and mounted, but no data transfer.

		▪ Flicker green: Data transfer in progress
9	Mgmt port (OOBM Port) with Activity/Link LED	Without an active network connection, this LED is off after power-on and self-test completes. With an active network connection, this LED operates as follows: ▪ Half-bright green: Port enabled and receiving Link indication from connected device. ▪ Flickering half-bright to full-bright green: Varying port activity level. ▪ Steady green: Port at high utilization.
10	Serial console port (RJ-45)	
11	USB Micro-B console port	
12	LED Mode button	Changes the behavior of the line module port LEDs. This button changes the LED behavior from the default Link/Activity behavior to cycle through the PoE, speed (Spd), and user (Usr) options.
13	UID (Unit Identification) LED	
14	PoE	Power-over-Ethernet
15	Chassis temperature status (Temp) LED	Indicates the status of the chassis temperature. If the temperature is at or below the specified rating, then the LED glows steady green,
16	Mgmt reset button	A recessed button that is used to reset the selected management module.
17	Mgmt state (Stby) LED	Indicates the status of the management module after booting. If the MM is the standby MM, then the LED glows steady green.

Power Supply LEDs

The Aruba 6400 has four power supply unit slots that support the Aruba X382 54DC 2700W AC power supply unit (PSU).

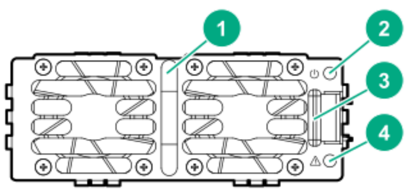
Figure 5 *Aruba X382 54DC 2700W AC Power Supply (JL372A)*



1	Power LED (green)

2	Power fail LED (amber)
3	Power supply handle
4	Latch release tab

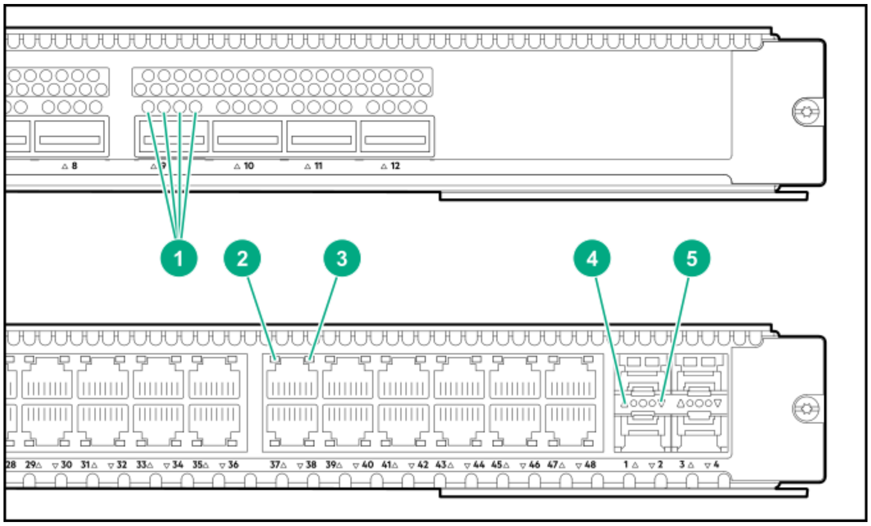
- A single PSU is sufficient for fans and management cards to come up and provide user access and diagnostics.
- At 220 V AC, only two PSUs are required for full operation and a single PSU is sufficient for the fans and management cards to come up and provide user access/diagnostics.
- At 220 V AC: Installing three PSUs offers 2+1 redundancy and installing all four PSUs offers 2+2 redundancy.
- At 110 V AC: The switch offers N + 1 redundancy.
- The PSUs are hot-swappable. The chassis can be connected to an AC power source for a given PSU slot while the PSU for that slot is being removed or installed.



1	Power LED (green)
2	Power fail LED (amber)
3	Power supply handle
4	Latch release tab

- A single PSU is sufficient for fans and management cards to come up and provide user access and diagnostics.
- At 220 V AC, only two PSUs are required for full operation and a single PSU is sufficient for the fans and management cards to come up and provide user access/diagnostics.
- At 220 V AC: Installing three PSUs offers 2+1 redundancy and installing all four PSUs offers 2+2 redundancy.
- At 110 V AC: The switch offers N + 1 redundancy.
- The PSUs are hot-swappable. The chassis can be connected to an AC power source for a given PSU slot while the PSU for that slot is being removed or installed.

Line module LEDs



1*	Line module 4-channel port LEDs
2 *	Line module port LED for upper port
3*	Line module port LED for lower port
4*	Line module port LED for upper uplink port
5*	Line module port LED for lower uplink port

Chapter 4 Boot commands

boot fabric-module

Syntax

`boot fabric-module <SLOT-ID>`

Description

Reboots the specified fabric module.

Command context

Manager (`#`)

Parameters

`<SLOT-ID>`

Specifies the member and slot of the module in the format `member/slot`. For example, to specify the module in member 1 slot 3, enter `1/3`.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The `boot fabric-module` command reboots the specified fabric module. Traffic performance is affected while the module is down.

If the specified module is the only fabric module in an up state, rebooting that module stops traffic switching between line modules and the line modules power down. The line modules power up when one fabric module returns to an up state.

This command is valid for fabric modules only.

Examples

Rebooting the fabric module in slot **1/3** when auto-confirm is not enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module.  Traffic performance may
be affected while the module is down.  Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n)? y

switch#
```

Rebooting the fabric module in slot **1/1** when auto-confirm is enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module.  Traffic performance may
be affected while the module is down.  Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n) y (auto-confirm)

switch#
```

boot line-module

Syntax

`boot line-module <SLOT-ID>`

Description

Reboots the specified line module.

Command context

Manager (`#`)

Parameters

<SLOT-ID>

Specifies the member and slot of the module in the format `member/slot`. For example, to specify the module in member 1 slot 3, enter `1/3`.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command is supported on switches that have multiple line modules.

Reboots the specified line module. Any traffic for the switch passing through the affected module (SSH, TELNET, and SNMP) is interrupted. It can take up to 2 minutes to reboot the module. During that time, you can monitor progress by viewing the event log.

This command is valid for line modules only.

Examples

Reloading the module in slot 1/1:

```
switch# boot line-module 1/1
This command will reboot the specified line module and interfaces on this
module will not send or receive packets while the module is down. Any
traffic passing through the line module will be interrupted. Management
sessions connected through the line module will be affected. It might take
up to 2 minutes to complete rebooting the module. During that time, you can
monitor progress by viewing the event log.
Do you want to continue (y/n)? y
switch#
```

boot management-module

Syntax

```
boot management-module {active | standby | <SLOT-ID>}
```

Description

Reboots the specified management module. Choose the management module to reboot by role (active or standby) or by slot number.

Command context

Manager (`#`)

Parameters

active

Selects the active management module.

standby

Selects the standby management module.

<SLOT-ID>

Specifies the member and slot of the management module in the format `member/slot`. For example, to specify the module in member 1 slot 5, enter `1/5`.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command is supported on switches that have multiple management modules.

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the `show images` command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the `boot` command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the `boot` command is aborted.

Hewlett Packarrd Enterprise recommends that you use the `boot management-module` command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the `boot management-module` command enables you to save the configuration, cancel the reboot, or both.

Examples

Rebooting the active management module when the standby management module is available:

```
switch# boot management-module active
The management-module in slot 1/5 is going down for reboot now.
```

Rebooting the active management module when the standby management module is not available:

```
switch# boot management-module 1/5
The management module in slot 1/5 is currently active and no
standby management module was found.
This will reboot the entire switch.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

boot set-default

Syntax

`boot set-default {primary | secondary}`

Description

Sets the default operating system image to use when the system is booted.

Command context

Manager (`#`)

Parameters

`primary`

Selects the primary network operating system image.

`secondary`

Selects the secondary network operating system image.

Authority

Administrators or local user group members with execution rights for this command.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

boot system

Syntax

`boot system [primary | secondary | serviceos]`

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Command context

Manager (`#`)

Parameters

`primary`
: Selects the primary operating system image for this reboot and sets the configured default operating system image to `primary` for future reboots.

`secondary`
: Selects the secondary operating system image for this reboot and sets the configured default operating system image to `secondary` for future reboots.

`serviceos`
: Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the ArubaOS-CX operating system and is used in rare cases when troubleshooting a switch.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the `show images` command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the `primary` or `secondary` optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

You can use the `boot set-default` command to change the configured default operating system image.

- If you select `serviceos` as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the `boot system` command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the `boot system` command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
```

```
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

show boot-history

Syntax

`show boot-history [all]`

Description

Shows boot information. When no parameters are specified, shows the most recent information about the boot operation, and the three previous boot operations for the active management module. When the `all` parameter is specified, shows the boot information for the active management module and all available line modules. To view boot-history on the standby, the command must be sent on the standby console.

Command context

Manager (`#`)

Parameters

`all`

Shows boot information for the active management module and all available line modules.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

`Index`

The position of the boot in the history file. Range: 0 to3.

`Boot ID`

A unique ID for the boot . A system-generated 128-bit string.

Current Boot, up for *<SECONDS>* seconds

For the current boot, the `show boot-history` command shows the number of seconds the module has been running on the current software.

Timestamp boot reason

For previous boot operations, the `show boot-history` command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values:

<DAEMON-NAME> **crash**

The daemon identified by *<DAEMON-NAME>* caused the module to boot.

Kernel crash

The operating system software associated with the module caused the module to boot.

Reboot requested through database

The reboot occurred because of a request made through the CLI or other API.

Uncontrolled reboot

The reason for the reboot is not known.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=================

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
switch#
```

Showing the boot history of the active management module and all line modules:

```
switch# show boot-history all
Management module
```

```
=================

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
```

bluetooth disable

Syntax

```
bluetooth disable
no bluetooth disable
```

Description

Disables the Bluetooth feature on the switch. The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE). Bluetooth is enabled by default.

The `no` form of this command enables the Bluetooth feature on the switch.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Example

Disabling Bluetooth on the switch. *<XXXX>* is the switch platform and *<NNNNNNNNNN>* is the device identifier.

```
switch(config)# bluetooth disable
switch# show bluetooth
Enabled              : No
Device name          : <XXXX>-<NNNNNNNNNN>

switch(config)# show running-config
...
bluetooth disabled
...
```

bluetooth enable

Syntax

```
bluetooth enable
no bluetooth enable
```

Description

This command enables the Bluetooth feature on the switch. The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE).

Default: Bluetooth is enabled by default.

The `no` form of this command disables the Bluetooth feature on the switch.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Usage

The default configuration of the Bluetooth feature is `enabled`. The output of the `show running-config` command includes Bluetooth information only if the Bluetooth feature is disabled.

The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE).

The Bluetooth feature requires the USB feature to be enabled. If the USB feature has been disabled, you must enable the USB feature before you can enable the Bluetooth feature.

Examples

```
switch(config)# bluetooth enable
```

clear events

Syntax

`clear events`

Description

Clears up event logs. Using the `show events` command will only display the logs generated after the `clear events` command.

Command context

Manager (`#`)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing all generated event logs:

```
switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2018-10-14:06:57:53.534384|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 27
2018-10-14:06:58:30.805504|lldpd|103|LOG_INFO|MSTR|1|Configured LLDP tx-timer to 36
2018-10-14:07:01:01.577564|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 49

switch# clear events

switch# show events
---------------------------------------------------
```

```
show event logs
---------------------------------------------------
2018-10-14:07:03:05.637544|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 34
```

clear ip errors

Syntax

`clear ip errors`

Description

Clears all IP error statistics.

Command context

Manager (`#`)

Authority

Administrators or local user group members with execution rights for this command.

Example

Clearing and showing ip errors:

```
switch# clear ip errors
switch# show ip errors
----------------------------------
Drop reason                Packets
----------------------------------
Malformed packets                0
IP address errors                0
...
```

domain-name

Syntax

```
domain-name <NAME>
no domain-name [<NAME>]
```

Description

Specifies the domain name of the switch.

The `no` form of this command sets the domain name to the default, which is no domain name.

Command context

`config`

Parameters

<NAME>

Specifies the domain name to be assigned to the switch. The first character of the name must be a letter or a number. Length: 1 to 32 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting and showing the domain name:

```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Setting the domain name to the default value:

```
switch(config)# no domain-name
switch(config)# show domain-name

switch(config)#
```

hostname

Syntax

```
hostname <HOSTNAME>
no hostname [<HOSTNAME>]
```

Description

Sets the host name of the switch.

The `no` form of this command sets the host name to the default value, which is `switch`.

Command context

`config`

Parameters

<HOSTNAME>

Specifies the host name. The first character of the host name must be a letter or a number. Length: 1 to 32 characters. Default: `switch`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
```

```
myswitch(config)# show hostname
myswitch
```

Setting the host name to the default value:

```
myswitch(config)# no hostname
switch(config)#
```

led locator

Syntax

`led locator {on | off | flashing}`

Description

Sets the state of the locator LED to `on`, `off` (default), or `flashing`.

Command context

Manager (`#`)

Parameters

`on`

Turns on the LED.

`off`

Turns off the LED, which is the default value.

`flashing`

Sets the LED to blink on and off repeatedly.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the state of the locator LED:

```
switch# led locator flashing
```

module admin-state

Syntax

`module <SLOT-ID> admin-state {diagnostic | down | up}`

Not supported on the 6300 Switch Series.

Description

Sets the administrative state of the specified line module.

Command context

config

Parameters

<SLOT-ID>

Specifies the member and slot of the module. For example, to specify the module in member 1, slot 3, enter the following:

`1/3`

`admin-state {diagnostic | down | up}`

Selects the administrative state in which to put the specified module:

`diagnostic`

Selects the `diagnostic` administrative state. Network traffic does not pass through the module.

`down`

Selects the `down` administrative state. Network traffic does not pass through the module.

`up`

Selects the `up` administrative state. The line module is fully operational. The `up` state is the default administrative state.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the administrative state of the module in slot **1/3** to `down`:

```
switch(config)# module 1/3 admin-state down
```

module product-number

Syntax

```
module <SLOT-ID> product-number [<PRODUCT-NUM>]
no module <SLOT-ID>
```

Not supported on the 6300 Switch Series.

Description

Changes the configuration of the switch to indicate that the specified member and slot number contains, or will contain, a line module.

The `no` form of this command removes the line module and its interfaces from the configuration. If there is a line module installed in the slot, the line module is powered off and then powered on.

Command context

`config`

Parameters

<SLOT-ID>

Specifies the member and slot in the form `m/s`, where `m` is the member number, and `s` is the slot number.

<PRODUCT-NUM>

Specifies the product number of the line module. For example: `JL363A`

If there is a line module installed in the slot when you execute this command, *<PRODUCT-NUM>* is optional. The switch reads the product number information from the module that is installed in the slot.

If there is no line module installed in the slot when you execute this command, *<PRODUCT-NUM>* is required.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The default configuration associated with a line module slot is:

- There is no module product number or interface configuration information associated with the slot. The slot is available for the installation with any supported line module.
- The Admin State is Up (which is the default value for Admin State).

To add a line module to the configuration, you must use the `module` command either before or after you install the physical module.

If you execute the `module` command after you install a line module in an empty slot, you can omit the *<PRODUCT-NUM>* variable. The switch reads the product information from the installed module.

If the module is not installed in the slot when you execute the module command, you must specify a value for the *<PRODUCT-NUM>* variable:

- The switch validates the product number of the module against the slot number you specify to ensure that the right type of module is configured for the specified slot.

  For example, the switch returns an error if you specify the product number of a line module for a slot reserved for management modules.
- You can configure the line module interfaces before the line module is installed.

When you install the physical line module in a preconfigured slot, the following actions occur:

- If a product number was specified in the command and it matches the product number of the installed module, the switch initializes the module.
- If a product number was specified in the command and the product number of the module does not match what was specified, the module device initialization fails.

The `no` form of the command removes the line module and its interfaces from the configuration and restores the line module slot to the default configuration.

If there is a line module installed in the slot when you execute the `no` form of the command, the command also powers off and then powers on the module. Traffic passing through the line module is stopped. Management sessions connected through the line module are also affected.

If the slot associated with the line module is in the default configuration, you can remove the module from the chassis without disrupting the operation of the switch.

Examples

Configuring slot 1/1 for future installation of a line module:

```
switch(config)# module 1/1 product-number jl363a
```

Configuring a line module that is already installed in slot 1/1:

```
switch(config)# module 1/1 product-number
```

Attempting to configure slot 1/1 for the future installation of a line module without specifying the product number (returned error shown):

```
switch(config)# module 1/1 product-number
Line module '1/4' is not physically available.  Please provide the product
number to preconfigure the line module.
```

Removing a module from the configuration:

```
switch(config)# no module 1/1
This command will power cycle the specified line module and restore its default
configuration. Any traffic passing through the line module will be interrupted.
Management sessions connected through the line module will be affected. It
might take a few minutes to complete this operation.

Do you want to continue (y/n)? y
switch(config)#
```

mtrace

Syntax

```
mtrace <IPV4-SRC-ADDR> <IPV4-GROUP-ADDR> [lhr <IPV4-LHR-ADDR>] [ttl <HOPS>]
   [vrf <VRF-NAME>]
```

Description

Traces the specified IPv4 source and group addresses.

Command context

Manager (#)

Parameters

IPV4-SRC-ADDR

Specifies the source IPv4 address to trace.

IPV4-GROUP-ADDR

Specifies the group IPv4 address to trace.

lhr *<IPV4-LHR-ADDR>*

Specifies the last hop router address from which to start the trace.

ttl *<HOPS>*

Specifies the Time-To-Live duration in hops. Range: 1 to 255 hops. Default: 8 hops.

vrf *<VRF-NAME>*

Specifies the name of the VRF. If a name is not specified the default VRF will be used.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Tracing with source, group, and LHR addresses and TTL:

```
(switch)# mtrace 20.0.0.1 239.1.1.1 lhr 10.1.1.1 ttl 10

Type escape sequence to abort.
Mtrace from 10.0.0.1 for Source 20.0.0.1 via Group 239.1.1.1
From destination(?) to source (?)...
Querying ful reverse path...
0  10.0.0.1
-1  30.0.0.1 PIM  0 ms
-2  40.0.0.1 PIM  2 ms
-3  50.0.0.1 PIM  100 ms
-4  60.0.0.1 PIM  156 ms
-5  20.0.0.1 PIM  123 ms
```

Tracing with source and group addresses:

```
(switch)# mtrace 200.0.0.1 239.1.1.1

Type escape sequence to abort.
Mtrace from self for Source 200.0.0.1 via Group 239.1.1.1
From destination(?) to source (?)...
Querying ful reverse path...
0  10.0.0.1
-1  30.0.0.1 PIM  0 ms
-2  40.0.0.1 PIM  2 ms
-3  50.0.0.1 PIM  100 ms
-4  60.0.0.1 PIM  156 ms
-5  200.0.0.1 PIM  123 ms
```

show bluetooth

Syntax

`show bluetooth`

Description

Shows general status information about the Bluetooth wireless management feature on the switch.

Command context

Operator (`>`) or Manager (`#`)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

This command shows status information about the following:

- The USB Bluetooth adapter
- Clients connected using Bluetooth
- The switch Bluetooth feature.

The output of the `show running-config` command includes Bluetooth information only if the Bluetooth feature is disabled.

The device name given to the switch includes the switch serial number to uniquely identify the switch while pairing with a mobile device.

The management IP address is a private network address created for managing the switch through a Bluetooth connection.

Examples

Example output when Bluetooth is enabled but no Bluetooth adapter is connected. *<XXXX>* is the switch platform and *<NNNNNNNNNN>* is the device identifier.

```
switch# show bluetooth
Enabled              : Yes
Device name          : <XXXX>-<NNNNNNNNNN>
Adapter State        : Absent
```

Example output when Bluetooth is enabled and there is a Bluetooth adapter connected:

```
switch# show bluetooth
Enabled              : Yes
Device name          : <XXXX>-<NNNNNNNNNN>
Adapter State        : Ready
Adapter IP address   : 192.168.99.1
Adapter MAC address  : 480fcf-af153a

Connected Clients
-----------------
Name             MAC Address      IP Address     Connected Since
--------------   --------------   ------------   ------------------------
Mark's iPhone    089734-b12000    192.168.99.10   2018-07-09 08:47:22 PDT
```

Example output when Bluetooth is disabled:

```
switch# show bluetooth
Enabled              : No
Device name          : <XXXX>-<NNNNNNNNNN>
```

show boot-history

Syntax

`show boot-history [all]`

Description

Shows boot information. When no parameters are specified, shows the most recent information about the boot operation, and the three previous boot operations for the active management module. When the `all` parameter is specified, shows the boot information for the active management module and all available line modules. To view boot-history on the standby, the command must be sent on the standby console.

Command context

Manager (`#`)

Parameters

`all`

Shows boot information for the active management module and all available line modules.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

`Index`

The position of the boot in the history file. Range: 0 to3.

`Boot ID`

A unique ID for the boot . A system-generated 128-bit string.

Current Boot, up for *<SECONDS>* seconds

For the current boot, the `show boot-history` command shows the number of seconds the module has been running on the current software.

Timestamp boot reason

For previous boot operations, the `show boot-history` command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values:

<DAEMON-NAME> **crash**

The daemon identified by *<DAEMON-NAME>* caused the module to boot.

Kernel crash

The operating system software associated with the module caused the module to boot.

Reboot requested through database

The reboot occurred because of a request made through the CLI or other API.

Uncontrolled reboot

The reason for the reboot is not known.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=================

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
switch#
```

Showing the boot history of the active management module and all line modules:

```
switch# show boot-history all
Management module
=================

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
```

show capacities

Syntax

`show capacities <FEATURE> [vsx-peer]`

Description

Shows system capacities and their values for all features or a specific feature.

Command context

Manager (`#`)

Parameters

<FEATURE>

Specifies a feature. For example, `aaa` or `vrrp`.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing all available capacities for BGP:

```
switch# show capacities bgp

System Capacities: Filter BGP
Capacities Name                                                              Value
----------------------------------------------------------------------------------
Maximum number of AS numbers in as-path attribute                               32
...
```

Showing all available capacities for mirroring:

```
switch# show capacities mirroring

System Capacities: Filter Mirroring
Capacities Name                                                              Value
----------------------------------------------------------------------------------
Maximum number of Mirror Sessions configurable in a system                       4
Maximum number of enabled Mirror Sessions in a system                            4
```

Showing all available capacities for MSTP:

```
switch# show capacities mstp

System Capacities: Filter MSTP
Capacities Name                                                              Value
----------------------------------------------------------------------------------
Maximum number of mstp instances configurable in a system                       64
```

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count

System Capacities: Filter VLAN Count
Capacities Name                                                              Value
----------------------------------------------------------------------------------
Maximum number of VLANs supported in the system                               4094
```

show capacities-status

Syntax

`show capacities-status <FEATURE> [vsx-peer]`

Description

Shows system capacities status and their values for all features or a specific feature.

Command context

Manager (`#`)

Parameters

<FEATURE>

Specifies the feature, for example `aaa` or `vrrp` for which to display capacities, values, and status. Required.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the system capacities status for all features:

```
switch# show capacities-status

System Capacities Status
Capacities Status Name                                                  Value Maximum
-------------------------------------------------------------------------------------
Number of active gateway mac addresses in a system                          0      16
Number of aspath-lists configured                                           0      64
Number of community-lists configured                                        0      64
...
```

Showing the system capacities status for BGP:

```
switch# show capacities-status bgp
System Capacities Status: Filter BGP
Capacities Status Name                                                  Value Maximum
-------------------------------------------------------------------------------------
Number of aspath-lists configured                                           0      64
Number of community-lists configured                                        0      64
Number of neighbors configured across all VRFs                              0      50
Number of peer groups configured across all VRFs                            0      25
Number of prefix-lists configured                                           0      64
Number of route-maps configured                                             0      64
Number of routes in BGP RIB                                                 0  256000
Number of route reflector clients configured across all VRFs                0      16
```

show core-dump

Syntax

`show core-dump` [all | *<SLOT-ID>*]

Description

Shows core dump information about the specified module. When no parameters are specified, shows only the core dumps generated in the current boot of the management module. When the `all` parameter is specified, shows all available core dumps.

Command context

Manager (`#`)

Parameters

`all`

Shows all available core dumps.

`<SLOT-ID>`

Shows the core dumps for the management module or line module in *`<SLOT-ID>`*. *`<SLOT-ID>`* specifies a physical location on the switch. Use the format `member/slot/port` (for example, `1/3/1`) for line modules. Use the format `member/slot` for management modules.

You must specify the slot ID for either the active management module, or the line module.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When no parameters are specified, the `show core-dump` command shows only the core dumps generated in the current boot of the management module. You can use this command to determine when any crashes are occurring in the current boot.

If no core dumps have occurred, the following message is displayed: `No core dumps are present`

To show core dump information for the standby management module, you must use the `standby` command to switch to the standby management module and then execute the `show core-dump` command.

In the output, the meaning of the information is the following:

`Daemon Name`

Identifies name of the daemon for which there is dump information.

`Instance ID`

Identifies the specific instance of the daemon shown in the `Daemon Name` column.

`Present`

Indicates the status of the core dump:

`Yes`

The core dump has completed and available for copying.

`In Progress`

Core dump generation is in progress. Do not attempt to copy this core dump.

`Timestamp`

Indicates the time the daemon crash occurred. The time is the local time using the time zone configured on the switch.

`Build ID`

Identifies additional information about the software image associated with the daemon.

Examples

Showing core dump information for the current boot of the active management module only:

```
switch# show core-dump
===================================================================================
Daemon Name      | Instance ID | Present     | Timestamp             | Build ID
===================================================================================
hpe-fand          1399          Yes          2017-08-04 19:05:34      1246d2a
hpe-sysmond       957           Yes          2017-08-04 19:05:29      1246d2a
===================================================================================
Total number of core dumps : 2
===================================================================================
```

Showing all core dumps:

```
switch# show core-dump all
===============================================================================
Management Module core-dumps
===============================================================================
Daemon Name      | Instance ID | Present     | Timestamp           | Build ID
===============================================================================
hpe-sysmond       513           Yes          2017-07-31 13:58:05    e70f101
hpe-tempd         1048          Yes          2017-08-13 13:31:53    e70f101
hpe-tempd         1052          Yes          2017-08-13 13:41:44    e70f101

Line Module core-dumps
===============================================================================
Line Module : 1/1
===============================================================================
dune_agent_0      18958         Yes          2017-08-12 11:50:17    e70f101
dune_agent_0      18842         Yes          2017-08-12 11:50:09    e70f101
===============================================================================
Total number of core dumps : 5
===============================================================================
```

show domain-name

Syntax

`show domain-name [vsx-peer]`

Description

Shows the current domain name.

Command context

Manager (#)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If there is no domain name configured, the CLI displays a blank line.

Example

Setting and showing the domain name:

```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
```

```
example.com
switch(config)#
```

show environment fan

Syntax

`show environment fan [vsf <MEMBER-ID> | vsx-peer]`

Description

Shows the status information for all fans and fan trays (if present) in the system.

Command context

Manager (`#`)

Parameters

`vsf <MEMBER-ID>`

Shows output from the VSF member-id on switches that support VSF.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

For fan trays, `Status` is one of the following values:

`ready`

The fan tray is operating normally.

`fault`

The fan tray is in a fault event. The status of the fan tray does not indicate the status of fans.

`empty`

The fan tray is not installed in the system.

For fans:

Speed

Indicates the relative speed of the fan based on the nominal speed range of the fan. Values are:

`Slow`

The fan is running at less than 25% of its maximum speed.

`Normal`

The fan is running at 25-49% of its maximum speed.

`Medium`

The fan is running at 50-74% of its maximum speed.

`Fast`

The fan is running at 75-99% of its maximum speed.

`Max`

The fan is running at 100% of its maximum speed.

`N/A`

The fan is not installed.

Direction

The direction of airflow through the fan. Values are:

`front-to-back`

Air flows from the front of the system to the back of the system.

`N/A`

The fan is not installed.

Status

Fan status. Values are:

`uninitialized`

The fan has not completed initialization.

`ok`

The fan is operating normally.

`fault`

The fan is in a fault state.

`empty`

The fan is not installed.

Examples

Showing output for systems with fan trays for 6400 switch series:

```
switch# show environment fan
Fan tray information
------------------------------------------------------------------------------
Mbr/Tray  Description                                  Status  Serial Number  Fans
------------------------------------------------------------------------------
1/1       R0X32A Aruba 6400 Fan Tray                   ready   SG9ZKJL7JW     4
1/2       R0X32A Aruba 6400 Fan Tray                   ready   SG9ZKJL7GL     4
1/3       R0X32A Aruba 6400 Fan Tray                   ready   SG9ZKJL78L     4
1/4       R0X32A Aruba 6400 Fan Tray                   ready   SG9ZKJL7GJ     4
Fan information
---------------------------------------------------------------------------
Mbr/Tray/Fan  Product  Serial Number  Speed   Direction      Status  RPM
              Name
---------------------------------------------------------------------------
1/1/1         N/A      N/A            slow    front-to-back  ok      5371
1/1/2         N/A      N/A            slow    front-to-back  ok      5320
1/1/3         N/A      N/A            slow    front-to-back  ok      5328
1/1/4         N/A      N/A            slow    front-to-back  ok      5256
1/2/1         N/A      N/A            slow    front-to-back  ok      5371
1/2/2         N/A      N/A            slow    front-to-back  ok      5349
1/2/3         N/A      N/A            slow    front-to-back  ok      5292
1/2/4         N/A      N/A            slow    front-to-back  ok      5349
1/3/1         N/A      N/A            slow    front-to-back  ok      5313
1/3/2         N/A      N/A            slow    front-to-back  ok      5371
1/3/3         N/A      N/A            slow    front-to-back  ok      5379
1/3/4         N/A      N/A            slow    front-to-back  ok      5379
1/4/1         N/A      N/A            slow    front-to-back  ok      5313
1/4/2         N/A      N/A            slow    front-to-back  ok      5299
1/4/3         N/A      N/A            slow    front-to-back  ok      5285
1/4/4         N/A      N/A            slow    front-to-back  ok      5371
```

Showing output for a system without a fan tray:

```
switch# show environment fan

Fan information
---------------------------------------------------------------
Fan   Serial Number  Speed   Direction      Status        RPM
---------------------------------------------------------------
1     SGXXXXXXXXXX   slow    front-to-back  ok            6000
2     SGXXXXXXXXXX   normal  front-to-back  ok            8000
3     SGXXXXXXXXXX   medium  front-to-back  ok            11000
4     SGXXXXXXXXXX   fast    front-to-back  ok            14000
5     SGXXXXXXXXXX   max     front-to-back  fault         16500
6     N/A            N/A     N/A            empty
...
```

show environment led

Syntax

`show environment led [vsf <MEMBER-ID>| vsx-peer]`

Description

Shows state and status information for all the configurable LEDs in the system.

Command context

Operator (`>`) or Manager (`#`)

Parameters

`vsf <MEMBER-ID>`

Shows output from the specified VSF member-id on switches that support VSF.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Show state and status for the environment LED:

```
switch# show environment led
Mbr/Name       State     Status
-------------------------------
1/locator      off       ok
```

show environment power-consumption

Syntax

`show environment power-consumption [vsx-peer]`

Not supported on the 6300 Switch Series.

Description

Shows the power being consumed by each management module, line card, and fabric card subsystem, and shows power consumption for the entire chassis.

Command context

Operator (`>`) or Manager (`#`)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

This command is only applicable to systems that support power consumption readings.

The power consumption values are updated once every minute.

The output of this command includes the following information:

Name

Shows the member number and slot number of the management module, line module, or fabric card module.

Type

Shows the type of module installed at the location specified by `Name`.

Description

Shows the product name and brief description of the module.

Usage

Shows the instantaneous power consumption of the module. Power consumption is shown in Watts.

Module Total Power Usage

Shows the total power consumption of all the modules listed. Power consumption is shown in Watts.

Chassis Total Power Usage

Shows the total instantaneous power consumed by the entire chassis, including modules and components that do not support individual power reporting. Power consumption is shown in Watts.

Chassis Total Power Available

Shows the total amount of power, in Watts, that can be supplied to the chassis.

Chassis Total Power Allocated

Shows total power, in Watts, that is allocated to powering the chassis and its installed modules.

Chassis Total Power Unallocated

Shows the total amount of power, in Watts, that has not been allocated to powering the chassis or its installed modules. This power can be used for additional hardware you install in the chassis.

Example

Showing the power consumption for an Aruba 6400 switch:

```
switch> show environment power-consumption
                                                                         Power
Name     Type                Description                                 Usage
-------------------------------------------------------------------------------
1/1      management-module   R0X31A 6400 Management Module               18 W
1/2      management-module                                               0 W
1/3      line-card-module                                                0 W
1/4      line-card-module    R0X39A 6400 48p 1GbE CL4 PoE 4SFP56 Mod     54 W
1/5      line-card-module                                                0 W
1/6      line-card-module    R0X39A 6400 48p 1GbE CL4 PoE 4SFP56 Mod     56 W
1/7      line-card-module    R0X39A 6400 48p 1GbE CL4 PoE 4SFP56 Mod     51 W
1/1      fabric-card-module  R0X24A 6405 Chassis                         71 W


Module Total Power Usage                                                 250 W
Chassis Total Power Usage                                                294 W

Chassis Total Power Available                                            1800 W
```

show environment power-supply

Syntax

`show environment power-supply [vsf <MEMBER-ID> | vsx-peer]`

Description

Shows status information about all power supplies in the switch.

Command context

Operator (`>`) or Manager (`#`)

Parameters

`vsf <MEMBER-ID>`

Shows output from the VSF member-id on switches that support VSF.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

The following information is provided for each power supply:

Mbr/PSU

Shows the member and slot number of the power supply.

Product Number

Shows the product number of the power supply.

Serial Number

Shows the serial number of the power supply, which uniquely identifies the power supply.

PSU Status

The status of the power supply. Values are:

`OK`

Power supply is operating normally.

`OK*`

Power supply is operating normally, but it is the only power supply in the chassis. One power supply is not sufficient to supply full power to the switch. When this value is shown, the output of the command also shows a message at the end of the displayed data.

`Absent`

No power supply is installed in the specified slot.

`Input fault`

The power supply has a fault condition on its input.

`Output fault`

The power supply has a fault condition on its output.

`Warning`

The power supply is not operating normally.

Wattage Maximum

Shows the maximum amount of wattage that the power supply can provide.

Example

Showing the output when only one power supply is installed in an Aruba 6400 switch chassis:

```
switch# show environment power-supply
         Product  Serial                 PSU           Wattage
Mbr/PSU  Number   Number                 Status        Maximum
--------------------------------------------------------------
1/1      R0X36A   CN91KMM2H3             OK            3000
1/2      N/A      N/A                    Absent        0
1/3      N/A      N/A                    Absent        0
1/4      N/A      N/A                    Absent        0
```

show environment rear-display-module

Syntax

`show environment rear-display-module [vsx-peer]`

Description

Shows information about the display module on the back of the switch (Aruba 8400 switches only).

Command context

Operator (>) or Manager (#)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing the rear display module information on the back of the switch:

```
switch> show environment rear-display-module

Rear display module is ready
Description: 8400 Rear Display Mod
Full Description: 8400 Rear Display Module
Serial number: SG00000000
Part number: 5300_0272
```

show environment temperature

Syntax

`show environment temperature [detail] [vsf <MEMBER-ID> | vsx-peer]`

Description

Shows the temperature information from sensors in the switch that affect fan control.

Command context

Operator (>) or Manager (#)

Parameters

`detail`

Shows detailed information from each temperature sensor.

`vsf`

Shows output from the VSF member-id on switches that support VSF.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Temperatures are shown in Celsius.

Valid values for status are the following:

`normal`

Sensor is within nominal temperature range.

`min`

Lowest temperature from this sensor.

`max`

Highest temperature from this sensor.

`low_critical`

Lowest threshold temperature for this sensor.

`critical`

Highest threshold temperature for this sensor.

`fault`

Fault event for this sensor.

`emergency`

Over temperature event for this sensor.

Examples

Showing current temperature information for a 6300 switch:

```
switch# show environment temperature
Temperature information
------------------------------------------------------------------------------
                                                     Current
Mbr/Slot-Sensor                 Module Type        temperature  Status
------------------------------------------------------------------------------
1/1-PHY-01-04                   line-card-module     45.00 C    normal
1/1-PHY-05-08                   line-card-module     45.00 C    normal
1/1-PHY-09-12                   line-card-module     46.00 C    normal
1/1-PHY-13-16                   line-card-module     47.00 C    normal
1/1-PHY-17-20                   line-card-module     47.00 C    normal
1/1-PHY-21-24                   line-card-module     50.00 C    normal
1/1-PHY-25-28                   line-card-module     45.00 C    normal
1/1-PHY-29-32                   line-card-module     47.00 C    normal
1/1-PHY-33-36                   line-card-module     48.00 C    normal
1/1-PHY-37-40                   line-card-module     47.00 C    normal
1/1-PHY-41-44                   line-card-module     48.00 C    normal
1/1-PHY-45-48                   line-card-module     49.00 C    normal
1/1-Switch-ASIC-Internal        line-card-module     56.25 C    normal

1/1-CPU-Zone-0                  management-module    50.00 C    normal
1/1-CPU-Zone-1                  management-module    50.00 C    normal
1/1-CPU-Zone-2                  management-module    50.00 C    normal
1/1-CPU-Zone-3                  management-module    51.00 C    normal
1/1-CPU-Zone-4                  management-module    51.00 C    normal
1/1-CPU-diode                   management-module    53.12 C    normal
1/1-DDR                         management-module    45.25 C    normal
1/1-Inlet-Air                   management-module    24.88 C    normal
1/1-MB-IBC                      management-module    45.62 C    normal
1/1-Switch-ASIC-diode           management-module    58.06 C    normal
```

Showing detailed temperature information for a 6300 switch:

```
switch# show environment temperature detail
Detailed temperature information
---------------------------------------------------------------
Mbr/Slot-Sensor      : 1/1-PHY-01-04
Module Type          : line-card-module
Module Description   : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status               : normal
Fan-state            : normal
Current temperature  : 45.00 C
Minimum temperature  : 41.00 C
Maximum temperature  : 50.00 C

Mbr/Slot-Sensor      : 1/1-PHY-05-08
Module Type          : line-card-module
Module Description   : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status               : normal
```

```
Fan-state               : normal
Current temperature     : 45.00 C
Minimum temperature     : 41.00 C
Maximum temperature     : 50.00 C

...
```

show events

Syntax

```
show events [ -e <EVENT-ID> |
     -s {alert | crit | debug | emer | err | info | notice | warn} |
     -r | -a | -i  <MEMBER/SLOT> | -n <count> |
     -m {active | standby} |
     -c {lldp | ospf | ... | } |
     -d {lldpd | hpe-fand | ... |}]
```

For 6300 switches:

```
show events [ -e <EVENT-ID> |
     -s {alert | crit | debug | emer | err | info | notice | warn} |
     -r | -a | -i  <MEMBER-SLOT> | -n <count> |
     -m {master | standby} |
     -c {lldp | ospf | ... | } |
     -d {lldpd | hpe-fand | ... |}]
```

Description

Shows event logs generated by the switch modules since the last reboot.

Command context

Manager (`#`)

Parameters

`-e` *`<EVENT-ID>`*

Shows the event logs for the specified event ID. Event ID range: 101 through 99999.

`-s {alert | crit | debug | emer | err | info | notice | warn}`

Shows the event logs for the specified severity. Select the severity from the following list:

- `alert`: Displays event logs with severity alert and above.
- `crit`: Displays event logs with severity critical and above.
- `debug`: Displays event logs with all severities.
- `emer`: Displays event logs with severity emergency only.
- `err`: Displays event logs with severity error and above.
- `info`: Displays event logs with severity info and above.
- `notice`: Displays event logs with severity notice and above.
- `warn`: Displays event logs with severity warning and above.

`-r`

Shows the most recent event logs first.

`-a`

Shows all event logs, including those events from previous boots.

`-i` *`<MEMBER-SLOT>`*

Shows the event logs for the specified slot ID on a 6400 switch.

-i *<MEMBER-ID>*

Shows the event logs for the specified VSF member ID on a 6300 switch.

-n *<count>*

Displays the specified number of event logs.

`-m {active | standby}`

Shows the event logs for the specified management card role on an 8400 or 6400 switch. Selecting *active* displays the event log for the AMM management card role and *standby* displays event logs for the SMM management card role.

`-m {master | standby}`

Shows the event logs for the specified role on a 6200 or 6300 switch. Selecting *master* displays the event log for the VSF master role and *standby* displays event logs for the VSF standby role.

`-c {lldp | ospf | ... | }`

Shows the event logs for the specified event category. Enter `show event -c` for a full listing of supported categories with descriptions.

`-d {lldpd | hpe-fand | ... |}`

Shows the event logs for the specified process. Enter `show event -d` for a full listing of supported daemons with descriptions.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Examples

Showing event logs:

```
switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for
bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in
Hardware
```

Showing the most recent event logs first:

```
switch# show events -r
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in
Hardware
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for
bridge_normal interface
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
```

Showing all event logs:

```
switch# show events -a
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for
bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in
Hardware
```

Showing event logs related to the DHCP relay agent:

```
switch# show events -c dhcp-relay
2016-05-31:06:26:27.363923|hpe-relay|110001|LOG_INFO|DHCP Relay Enabled
2016-05-31:07:08:51.351755|hpe-relay|110002|LOG_INFO|DHCP Relay Disabled
```

Showing event logs related to the DHCPv6 relay agent:

```
switch# show events -c dhcpv6-relay
2016-05-31:06:26:27.363923|hpe-relay|109001|LOG_INFO|DHCPv6 Relay Enabled
2016-05-31:07:08:51.351755|hpe-relay|109002|LOG_INFO|DHCPv6 Relay Disabled
```

Showing event logs related to IRDP:

```
switch# switch# show events -c irdp
2016-05-31:06:26:27.363923|hpe-rdiscd|111001|LOG_INFO|IRDP enabled on interface %s
2016-05-31:07:08:51.351755|hpe-rdiscd|111002|LOG_INFO|IRDP disabled on interface %s
```

Showing event logs related to LACP:

```
switch# show events -c lacp
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
```

Showing event logs as per the specified management card role for a 6400 switch:

```
switch# show events -m active
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for
bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in
Hardware
```

Showing event logs as per the specified management card role for a switch:

```
switch# show events -m master
---------------------------------------------------
show event logs
---------------------------------------------------
2020-04-22T05:36:13.348594+00:00 6300 lldpd[3055]: Event|109|LOG_
INFO|MSTR|1|Configured LLDP tx-delay to 2
2020-04-22T05:36:14.430166+00:00 6300 vrfmgrd[3053]: Event|5401|LOG_
INFO|MSTR|1|Created a vrf entity b1721d27-41c2-485d-9bae-2cfcbc9bd13d
2020-04-22T05:36:14.942597+00:00 6300 vrfmgrd[3053]: Event|5401|LOG_
INFO|MSTR|1|Created a vrf entity 5eb532c9-5b4d-4d83-b34a-db24ae542d4e
```

Showing event logs as per the specified slot ID for a 6400 switch:

```
switch# show events -i 1/1
---------------------------------------------------
show event logs
---------------------------------------------------
2017-08-17:22:32:25.743991|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
utilization poll interval is changed to 313
2017-08-17:22:33:01.692860|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
utilization poll interval is changed to 23
2017-08-17:22:33:06.181436|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
utilization poll interval is changed to 512
2017-08-17:22:33:06.181436|systemd-coredump|1201|LOG_CRIT|LC|1/1|hpe-sysmond crashed
due to signal:11
```

Showing event logs as per the specified slot ID:

```
switch# show events -i 1
---------------------------------------------------
Event logs from current boot
---------------------------------------------------
2020-04-22T05:36:14.430166+00:00 6300 vrfmgrd[3053]: Event|5401|LOG_
INFO|MSTR|1|Created a vrf entity b1721d27-41c2-485d-9bae-2cfcbc9bd13d
2020-04-22T05:36:14.942597+00:00 6300 vrfmgrd[3053]: Event|5401|LOG_
INFO|MSTR|1|Created a vrf entity 5eb532c9-5b4d-4d83-b34a-db24ae542d4e
2020-04-22T05:36:15.886252+00:00 6300 vsfd[710]: Event|9903|LOG_INFO|MSTR|1|Master 1
boot complete
```

Showing event logs as per the specified process:

```
switch# show events -d lacpd
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
```

Displaying the specified number of event logs:

```
switch# show events -n 5
---------------------------------------------------
show event logs
```

```
--------------------------------------------------
2018-03-21:06:12:15.500603|arpmgrd|6101|LOG_INFO|AMM|-|ARPMGRD daemon has started
2018-03-21:06:12:17.734405|lldpd|109|LOG_INFO|AMM|-|Configured LLDP tx-delay to 2
2018-03-21:06:12:17.740517|lacpd|1307|LOG_INFO|AMM|-|LACP system ID set to
70:72:cf:d4:34:42
2018-03-21:06:12:17.743491|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity
42cc3df7-1113-412f-b5cb-e8227b8c22f2
2018-03-21:06:12:17.904008|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity
4409133e-2071-4ab8-adfe-f9662c06b889
```

show fabric

Syntax

`show fabric [<SLOT-ID>] [vsx-peer]`

Not supported on the 6300 Switch Series.

Description

Shows information about the installed fabrics.

Command context

Operator (`>`) or Manager (`#`)

Parameters

`<SLOT-ID>`

Specifies the member and slot of the fabric to show. For example, to show the module in member 1, slot 2, enter the following:

`1/2`

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing all fabrics on Aruba 6400 switches that have two fabrics:

```
switch# show fabric

Fabric Modules
==============

     Product                                          Serial
Name Number  Description                              Number     Status
---- ------- --------------------------------         ---------- ----------------
1/1  R0X25A  6410 Chassis                             SG9ZKM9999 Ready
1/2  R0X25A  6410 Chassis                             SG9ZKM9999 Ready
```

Showing all fabrics on Aruba 6400 switches that have one fabric:

```
switch# show fabric

Fabric Modules
==============

     Product                                        Serial
Name Number  Description                            Number     Status
---- ------- -------------------------------------- ---------- ----------------
1/1  R0X24A  6405 Chassis                           SG9ZKM9076 Ready
```

Showing a single fabric module on Aruba 6400 switches:

```
switch# show fabric 1/1

Fabric module 1/1 is ready
Admin state: Up
Description: 6405 Chassis
Full Description: 6405 Chassis
Serial number: SG00000000
Product number: R0X24A
```

show hostname

Syntax

`show hostname [vsx-peer]`

Description

Shows the current host name.

Command context

Manager (`#`)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

show images

Syntax

```
show images [vsx-peer]
```

Description

Shows information about the software in the primary and secondary images.

Command context

Operator (>) or Manager (#)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing the primary and secondary images on a 6300 switch:

```
switch(config)# show images
---------------------------------------------------------------------------
ArubaOS-CX Primary Image
---------------------------------------------------------------------------
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-22 17:00:46 PDT
SHA-256 : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa

---------------------------------------------------------------------------
ArubaOS-CX Secondary Image
---------------------------------------------------------------------------
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-22 17:00:46 PDT
SHA-256 : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa

Default Image : secondary

------------------------------------------------------
Management Module 1/1 (Active)
------------------------------------------------------
Active Image       : secondary
Service OS Version : FL.01.05.0001-internal
BIOS Version       : FL.01.0001
```

Showing the primary and secondary images on a 6400 switch:

```
switch(config)# show images
---------------------------------------------------------------------------
```

```
ArubaOS-CX Primary Image
---------------------------------------------------------------------------
Version : FL.xx.xx.xxxxQ-2710-gd4ac39f30c9
Size    : 766 MB
Date    : 2019-10-30 17:22:01 PDT
SHA-256 : e560ca9141f425d19024d122573c5ff730df2a9a726488212263b45ea00382cf

---------------------------------------------------------------------------
ArubaOS-CX Secondary Image
---------------------------------------------------------------------------
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-21 19:36:26 PDT
SHA-256 : 657e28adc1b512217ce780e3523c37c94db3d3420231deac1ab9aaa8324dc6b9

Default Image : secondary

------------------------------------------------------
Management Module 1/1 (Active)
------------------------------------------------------
Active Image       : secondary
Service OS Version : FL.01.05.0001-internal
BIOS Version       : FL.01.0001
```

show ip errors

Syntax

`show ip errors [vsx-peer]`

Description

Shows IP error statistics for packets received by the switch since the switch was last booted.

Command context

Operator (>) or Manager (#)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

IP error info about received packets is collected from each active line card on the switch and is preserved during failover events. Error counts are cleared when the switch is rebooted.

Drop reasons are the following:

Malformed packet

The packet does not conform to TCP/IP protocol standards such as packet length or internet header length.

A large number of malformed packets can indicate that there are hardware malfunctions such as loose cables, network card malfunctions, or that a DOS (denial of service) attack is occurring.

IP address error

The packet has an error in the destination or source IP address. Examples of IP address errors include the following:

- The source IP address and destination IP address are the same.
- There is no destination IP address.
- The source IP address is a multicast IP address.
- The forwarding header of an IPv6 address is empty.
- There is no source IP address for an IPv6 packet.

Invalid TTLs

The TTL (time to live) value of the packet reached zero. The packet was discarded because it traversed the maximum number of hops permitted by the TTL value.

TTLs are used to prevent packets from being circulated on the network endlessly.

Example

Showing ip error statistics for packets received by the switch:

```
switch# show ip errors
----------------------------------
Drop reason                Packets
----------------------------------
Malformed packets                1
IP address errors               10
...
```

show module

Syntax

`show module [<SLOT-ID>] [vsx-peer]`

Description

Shows information about installed line modules and management modules.

Command context

Operator (`>`) or Manager (`#`)

Parameters

<SLOT-ID>

Specifies the member and slot numbers in format `member/slot`. For example, to show the module in member 1, slot 3, enter `1/3`.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Identifies and shows status information about the line modules and management modules that are installed in the switch.

If you use the *<SLOT-ID>* parameter to specify a slot that does not have a line module installed, a message similar to the following example is displayed:

`Module 1/4 is not physically present.`

To show the configuration information—if any—associated with that line module slot, use the `show running-configuration` command.

Status is one of the following values:

`Active`
: This module is the active management module.

`Standby`
: This module is the standby management module.

`Deinitializing`
: The module is being deinitialized.

`Diagnostic`
: The module is in a state used for troubleshooting.

`Down`
: The module is physically present but is powered down.

`Empty`
: The module is not installed in the chassis.

`Failed`
: The module has experienced an error and failed.

`Failover`
: This module is a fabric module or a line module, and it is in the process of connecting to the new active management module during a management module failover event.

`Initializing`
: The module is being initialized.

`Present`
: The module hardware is installed in the chassis.

`Ready`
: The module is available for use.

`Updating`
: A firmware update is being applied to the module.

Examples

Showing all installed modules (Aruba 6300 switch):

```
switch(config)# show module

Management Modules
==================

     Product                                        Serial
Name Number  Description                            Number     Status
---- ------- -------------------------------------- ---------- ----------------
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch        ID9ZKHN090 Active (local)
```

```
Line Modules
============

     Product                                          Serial
Name Number  Description                              Number     Status
---- ------- ---------------------------------------- ---------- ----------------
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch          ID9ZKHN090 Ready
```

Showing a line module (Aruba 6400 switch):

```
switch# show module 1/3

Line module 1/3 is ready
 Admin state: Up
 Description: 6400 24p 10GT 4SFP56 Mod
 Full Description: 6400 24-port 10GBASE-T and 4-port SFP56 Module
 Serial number: SG9ZKMS045
 Product number: R0X42A
 Power priority: 128
```

Showing a slot that does not contain a line module:

```
switch(config)# show module 1/3
Module 1/3 is not physically present
```

show running-config

Syntax

`show running-config [<FEATURE>] [all] [vsx-peer]`

Description

Shows the current nondefault configuration running on the switch. No user information is displayed.

Command context

Manager (`#`)

Parameters

<FEATURE>

Specifies the name of a feature. For a list of feature names, enter the `show running-config` command, followed by a space, followed by a question mark (?). When the `json` parameter is used, the `vsx-peer` parameter is not applicable.

`all`

Shows all default values for the current running configuration.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the current running configuration:

```
switch> show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.XXXX
!
lldp enable
linecard-module LC1 part-number JL363A
vrf green
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!

vlan 1
    no shutdown
vlan 20
    no shutdown
vlan 30
    no shutdown
interface 1/1/1
    no shutdown
    no routing
    vlan access 30
interface 1/1/32
    no shutdown
    no routing
    vlan access 20
interface bridge_normal-1
    no shutdown
interface bridge_normal-2
    no shutdown
interface vlan20
   no shutdown
   vrf attach green
   ip address 20.0.0.44/24
   ip ospf 1 area 0.0.0.0
   ip pim-sparse enable

interface vlan30
   no shutdown
   vrf attach green
   ip address 30.0.0.44/24
   ip ospf 1 area 0.0.0.0
   ip pim-sparse enable

   ip pim-sparse hello-interval 100
```

Showing the current running configuration in json format:

```
switch> show running-config json
Running-configuration in JSON:
{
    "Monitoring_Policy_Script": {
        "system_resource_monitor_mm1.1.0": {
            "Monitoring_Policy_Instance": {
                "system_resource_monitor_mm1.1.0/system_resource_monitor_
mm1.1.0.default": {
                    "name": "system_resource_monitor_mm1.1.0.default",
                    "origin": "system",
                    "parameters_values": {
                        "long_term_high_threshold": "70",
                        "long_term_normal_threshold": "60",
                        "long_term_time_period": "480",
                        "medium_term_high_threshold": "80",
                        "medium_term_normal_threshold": "60",
                        "medium_term_time_period": "120",
                        "short_term_high_threshold": "90",
                        "short_term_normal_threshold": "80",
                        "short_term_time_period": "5"
                    }
                }
            },
...
...
...
...
```

Show the current running configuration without default values:

```
switch(config)# show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
!
!
!
!
!
vlan 1
switch(config)# show running-config all
Current configuration:
!
!Version ArubaOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
!
!
!
!
!
vlan 1
switch(config)#
```

Show the current running configuration with default values:

```
switch(config)# snmp-server vrf mgmt
switch(config)# show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
snmp-server vrf mgmt
!
!
!
!
!
vlan 1
switch(config)#
switch(config)#
switch(config)# show running-config all
Current configuration:
!
!Version ArubaOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
snmp-server vrf mgmt
snmp-server agent-port 161
snmp-server community public
!
!
!
!
!
vlan 1
switch(config)#
```

show running-config current-context

Syntax

`show running-config current-context`

Description

Shows the current non-default configuration running on the switch in the current command context.

Command context

`config` or a child of `config`. See Usage.

Authority

Administrators or local user group members with execution rights for this command.

Usage

You can enter this command from the following configuration contexts:

- Any child of the global configuration (`config`) context. If the child context has instances—such as interfaces—you can enter the command in the context of a specific instance.

  Support for this command is provided for one level below the `config` context. For example, entering this command for a child of a child of the `config` context not supported.

  If you enter the command on a child of the `config` context, the current configuration of that context and the children of that context are displayed.

- The global configuration (`config`) context.

  If you enter this command in the global configuration (`config`) context, it shows the running configuration of the entire switch. Use the `show running-configuration` command instead.

Examples

On the 6400 Switch Series, interface identification differs.

Showing the running configuration for the current interface:

```
switch(config-if)# show running-config current-context
interface 1/1/1
    vsx-sync qos vlans
    no shutdown
    description Example interface
vlan access 1
    exit
```

Showing the current running configuration for the management interface:

```
switch(config-if-mgmt)# show running-config current-context
interface mgmt
    no shutdown
    ip static 10.0.0.1/24
    default-gateway 10.0.0.8
    nameserver 10.0.0.1
```

Showing the running configuration for the external storage share named `nasfiles`:

```
switch(config-external-storage-nasfiles)# show running-config current-context
external-storage nasfiles
   address 192.168.0.1
   vrf default
   username nasuser
   password ciphertext AQBapalKj+XMsZumHEwIc9OR6YcOw5Z6Bh9rV+9ZtKDKzvbaBAAAAB1CTrM=
   type scp
   directory /home/nas
   enable
switch(config-external-storage-nasfiles)#
```

Showing the running configuration for a context that does not have instances:

```
switch(config-vsx)# show run current-context
vsx
    inter-switch-link 1/1/1
    role secondary
    vsx-sync sflow time
```

show startup-config

Syntax

```
show startup-config [json]
```

Description

Shows the contents of the startup configuration.

Switches in the `factory-default` configuration do not have a startup configuration to display.

Command context

Manager (#)

Parameters

json

Display output in JSON format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the startup-configuration in non-JSON format for a 6300 switch:

```
switch(config)# show startup-config
Startup configuration:
!
!Version ArubaOS-CX FL.xx.xx.xxxx
!export-password: default
hostname BLDG01-F1
user admin group administrators password ciphertext

AQBapWl8I2ZunZ43NE/8KlbQ7zYC4gTT6uSFYi6n6wyY9PdBYgAAACONCR/3+AcNvzRBch0DoG7W9z84LpJA
+6C9SKfNwCqi5/
nUPk/ZOvN91/EQXvPNkHtBtQWyYZqfkebbEH78VWRHfWZjApv4II9qmQfxpA79wEvzshdzZmuAKrm
user ateam group administrators password ciphertext

AQBapcPqMXoF+H10NKrqAedXLvlSRwf4wUEL22hXGD6ZBhicYgAAAGsbh70DKg1u+Ze1wxgmDXjkGO3bseYi
R3LKQg66vrfrqR/
M3oLlliPdZDnq9XMMvCL+7jBbYhYes8+uDxuSTh8kdkd/qj3lo5FUuC5fENgCjU0YI1l7qtU+YEnsj
!
!
!
!
radius-server host 10.10.10.15
!
radius dyn-authorization enable
ssh server vrf default
ssh server vrf mgmt
!
!
!
!
!
router ospf 1
```

```
    router-id 1.63.63.1
    area 0.0.0.0
vlan 1
vlan 66
    name vlan66
vlan 67
    name vlan67
vlan 999
    name vlan999
vlan 4000
spanning-tree
interface mgmt
    no shutdown
    ip static 10.6.9.15/24
    default-gateway 10.6.9.1
```

Showing the startup-configuration in JSON format:

```
switch# show startup-config json
Startup configuration:
{
    "AAA_Server_Group": {
        "local": {
            "group_name": "local"
        },
        "none": {
            "group_name": "none"
        }
    },
...
```

show system

Syntax

`show system [vsx-peer]`

Description

Shows general status information about the system.

Command context

Operator (`>`) or Manager (`#`)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

CPU utilization represents the average utilization across all the CPU cores.

System Contact, System Location, and System Description can be set with the `snmp-server` command.

Examples

Showing system information on a 6300 switch:

```
switch(config)# show system
Hostname           : switch
System Description : FL.10.xx.xxxxx
System Contact     :
System Location    :

Vendor             : Aruba
Product Name       : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Chassis Serial Nbr : ID9ZKHN090
Base MAC Address   : 9020c2-245080
ArubaOS-CX Version : FL.10.xx.xxxx

Time Zone          : UTC

Up Time            : 5 days, 15 hours, 33 minutes
CPU Util (%)       : 21
Memory Usage (%)   : 19
```

Showing system information om a 6400 switch:

```
switch(config)# show system
Hostname           : switch
System Description : FL.10.xx.xxxxx
System Contact     :
System Location    :

Vendor             : Aruba
Product Name       : R0X24A 6405 Chassis
Chassis Serial Nbr : SG9ZKM7206
Base MAC Address   : 9020c2-dc4700
ArubaOS-CX Version : FL.10.xx.xxxxx

Time Zone          : UTC

Up Time            : 32 minutes
CPU Util (%)       : 3
Memory Usage (%)   : 10
BLDG02-F3(config)#
```

show system error-counter-monitor

Syntax

`show system error-counter-monitor {basic <PORT-NUM> | extended} [vsx-peer]`

Description

Shows error counter statistics.

Command context

Manager (#)

Parameters

`basic <PORT-NUM>`

Specifies a physical port on the switch. Use the format `member/slot/port` (for example, `1/3/1`).

`extended`

Shows statistics for all interfaces.

Command context

Manager (`#`)

Examples

Showing error counter statistics for interface 1/1/1:

```
switch# show system error-counter-monitor basic 1/1/1

Interface error counter statistics for 1/1/1

Error Counter                       Value
-----------------------------------------
EtherStatsOversizePkts              983
EtherStatsUndersizePkts             1024
EtherStatsJabbers                   10
Dot3StatsAlignmentErrors            462
Dot3StatsFCSErrors                  321
Dot3StatsLateCollisions             2024
EtherStatsFragments                 121
Dot3StatsExcessiveCollisions        1025
IfInBroadcastPkts                   2001
```

Showing error counter statistics for all interfaces:

```
switch# show system error-counter-monitor extended

Interface error counter statistics for 1/1/1

Error Counter                       Value
-----------------------------------------
EtherStatsOversizePkts              983
EtherStatsUndersizePkts             1024
EtherStatsJabbers                   10
Dot3StatsAlignmentErrors            462
Dot3StatsFCSErrors                  321
Dot3StatsLateCollisions             2024
EtherStatsFragments                 121
Dot3StatsExcessiveCollisions        1025
IfInBroadcastPkts                   2001
...
...
Interface error counter statistics for 1/8/32

Error Counter                      Value
-----------------------------------------
EtherStatsOversizePkts              0
...
```

show system resource-utilization

Syntax

`show system resource-utilization [daemon <DAEMON-NAME> | module <SLOT-ID>] [vsx-peer]`

Description

Shows information about the usage of system resources such as CPU, memory, and open file descriptors.

Command context

Manager (`#`)

Parameters

daemon *<DAEMON-NAME>*

Shows the filtered resource utilization data for the process specified by *<DAEMON-NAME>* only.

For a list of daemons that log events, enter `show events -d ?` from a switch prompt in the manager (`#`) context.

module *<SLOT-ID>*

Shows the filtered resource utilization data for the line module specified by *<SLOT-ID>* only.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing all system resource utilization data:

```
switch# show system resource-utilization
System Resources:
Processes: 70
CPU usage(%): 20
Memory usage(%): 25
Open FD's: 1024

Process              CPU Usage(%)          Memory Usage(%)     Open FD's
-----------------------------------------------------------------------
pmd                     2                       1                14
hpe-sysmond             1                       2                11
hpe-mgmdd               0                       1                5
...
```

Showing the resource utilization data for the pmd process:

```
switch# show system resource-utilization daemon pmd
Process              CPU Usage          Memory Usage     Open FD's
-----------------------------------------------------------------------
pmd                      2                 1             14
```

Showing resource utilization data when system resource utilization polling is disabled:

```
switch# show system resource-utilization
System resource utilization data poll is currently disabled
```

Showing resource utilization data for a line module:

```
switch# show system resource-utilization module 1/1
System Resource utilization for line card module: 1/1
CPU usage(%): 0
Memory usage(%): 35
Open FD's: 512
```

show tech

Syntax

`show tech [basic | <FEATURE>] [local-file]`

Description

Shows detailed information about switch features by automatically running the `show` commands associated with the feature. If no parameters are specified, the `show tech` command shows information about all switch features. Technical support personnel use the output from this command for troubleshooting.

Command context

Manager (`#`)

Parameters

`basic`

Specifies showing a basic set of information.

`<FEATURE>`

Specifies the name of a feature. For a list of feature names, enter the `show tech` command, followed by a space, followed by a question mark (?).

`local-file`

Shows the output of the `show tech` command to a local text file.

Authority

Administrators or local user group members with execution rights for this command.

Usage

To terminate the output of the `show tech` command, enter Ctrl+C.

If the command was not terminated with Ctrl+C, at the end of the output, the `show tech` command shows the following:

- The time consumed to execute the command.
- The list of failed `show` commands, if any.

To get a copy of the local text file content created with the show tech command that is used with the local-file parameter, use the `copy show-tech local-file` command.

Example

Showing the basic set of system information:

```
switch# show tech basic
==============================================================
Show Tech executed on Wed Sep  6 16:50:37 2017
==============================================================
==============================================================
[Begin] Feature basic
==============================================================

*******************************
Command : show core-dump all
*******************************
no core dumps are present

...
==============================================================
[End] Feature basic
==============================================================

==============================================================
1 show tech command failed
==============================================================
Failed command:
  1. show boot-history
==============================================================
Show tech took 3.000000 seconds for execution
```

Directing the output of the **show tech basic** command to the local text file:

```
switch# show tech basic local-file
Show Tech output stored in local-file. Please use 'copy show-tech local-file'
to copy-out this file.
```

show usb

Syntax

`show usb [vsx-peer]`

Description

Shows the USB port configuration and mount settings.

Command context

Operator (>) or Manager (#)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

If USB has not been enabled:

```
switch> show usb
Enabled: No
Mounted: No
```

If USB has been enabled, but no device has been mounted:

```
switch> show usb
Enabled: Yes
Mounted: No
```

If USB has been enabled and a device mounted:

```
switch> show usb
Enabled: Yes
Mounted: Yes
```

show usb file-system

Syntax

`show usb file-system [<PATH>]`

Description

Shows directory listings for a mounted USB device. When entered without the <PATH> parameter the top level directory tree is shown.

Command context

Operator (>) or Manager (#)

Parameters

<PATH>

Specifies the file path to show. A leading "/" in the path is optional.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Adding a leading "/" as the first character of the *<PATH>* parameter is optional.

Attempting to enter '..' as any part of the *<PATH>* will generate an invalid path argument error. Only fully-qualified path names are supported.

Examples

Showing the top level directory tree:

```
switch# show usb file-system
/mnt/usb:
'System Volume Information'  dir1'

/mnt/usb/System Volume Information':
IndexerVolumeGuid  WPSettings.dat

/mnt/usb/dir1:
dir2  test1

/mnt/usb/dir1/dir2:
test2
```

Showing available path options from the top level:

```
switch# show usb file-system /
total 64
drwxrwxrwx 2 32768 Jan 22 16:27 'System Volume Information'
drwxrwxrwx 3 32768 Mar  5 15:26 dir1
```

Showing the contents of a specific folder:

```
switch# show usb file-system /dir1
total 32
drwxrwxrwx 2 32768 Mar  5 15:26 dir2
-rwxrwxrwx 1     0 Feb  5 18:08 test1

switch# show usb file-system dir1/dir2
total 0
-rwxrwxrwx 1 0 Feb  6 05:35 test2
```

Attempting to enter an invalid character in the path:

```
switch# show usb file-system dir1/../..
Invalid path argument
```

show version

Syntax

`show version [vsx-peer]`

Description

Shows version information about the network operating system software, service operating system software, and BIOS.

Command context

Operator (>) or Manager (#)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing version information for a 6300 switch:

```
switch(config)# show version
-----------------------------------------------------------------------------
ArubaOS-CX
(c) Copyright 2017-2020 Hewlett Packard Enterprise Development LP
-----------------------------------------------------------------------------
Version      : FL.xx.xx.xxxx
Build Date   : 2020-10-22 17:00:46 PDT
Build ID     : ArubaOS-CX:FL.xx.xx.xxxx:85c3c2f3d59e:201910222335
Build SHA    : 85c3c2f3d59ec8318ba97178fad387aecb671b33
Active Image : secondary

Service OS Version : FL.01.05.0001-internal
BIOS Version       : FL.01.0001
```

system resource-utilization poll-interval

Syntax

`system resource-utilization poll-interval <SECONDS>`

Description

Configures the polling interval for system resource information collection and recording such as CPU and memory usage.

Command context

`config`

Parameters

<SECONDS>

Specifies the poll interval in seconds. Range: 10-3600. Default: 10.

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring the system resource utilization poll interval:

```
switch(config)# system resource-utilization poll-interval 20
```

top cpu

Syntax

`top cpu`

Description

Shows CPU utilization information.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing top CPU information:

```
switch# top cpu
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks:  76 total, 2 running, 74 sleeping,  0 stopped,  0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total,  2487508 free,  897040 used,   661948 buff/cache
KiB Swap:       0 total,        0 free,       0 used,  2859196 avail Mem

  PID USER      PRI  NI   VIRT    RES   SHR S  %CPU %MEM      TIME+ COMMAND
...
```

top memory

Syntax

`top memory`

Description

Shows memory utilization information.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing top memory:

```
switch> top memory
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks:  76 total, 2 running, 74 sleeping,  0 stopped,  0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total,  2487508 free,  897040 used,   661948 buff/cache
```

```
KiB Swap:        0 total,        0 free,        0 used,  2859196 avail Mem

  PID USER      PRI  NI   VIRT    RES   SHR S  %CPU %MEM      TIME+ COMMAND
...
```

usb

Syntax

`usb`

`no usb`

Description

Enables the USB ports on the switch. This setting is persistent across switch reboots and management module failovers. Both active and standby management modules are affected by this setting.

The `no` form of this command disables the USB ports.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Example

Enabling USB ports:

```
switch(config)# usb
```

Disabling USB ports when a USB drive is mounted:

```
switch(config)# no usb
```

usb mount | unmount

Syntax

`usb {mount | unmount}`

Description

Enables or disables the inserted USB drive.

Command context

Manager (`#`)

Parameters

`mount`

Enables the inserted USB drive.

`unmount`

Disables the inserted USB drive in preparation for removal.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If USB has been enabled in the configuration, the USB port on the active management module is available for mounting a USB drive. The USB port on the standby management module is not available.

An inserted USB drive must be mounted each time the switch boots or fails over to a different management module.

A USB drive must be unmounted before removal.

The supported USB file systems are FAT16 and FAT32.

Examples

Mounting a USB drive in the USB port:

```
switch# usb mount
```

Unmounting a USB drive:

```
switch# usb unmount
```

Chapter 6 External storage

The switch has limited capacity to store data, collected by switch features and protocols. You can provide virtually unlimited storage capacity by adding user-supplied external storage volumes. Supported volume types and storage protocols include: NFSv3, NFSv4, and SCP (sshfs).

One application of external storage is the saving and restoring of DHCP lease files over SCP or NFS network attached storage systems. SCP file system protocol uses a user mode process to emulate a network file system. The key advantage is packet level encryption and simple configuration. The key disadvantage is slow performance.

You can set up external storage volume credentials and then enable it. A storage management process acts on your requests by enabling the storage volume using the requested storage protocol. You can disable the external storage volume or set it up but leave it disable.

The feature maintains storage volume state. The states are: *disabled* (down), *connecting* (establishing connection), *operational* (up), and *unaccessible* (unavailable).

If a storage volume is unavailable, the system attempts to reconnect periodically. Multiple volumes could connect concurrently. If one connection times out the others can connect immediately.

The system supports server connection through data and management ports.

Data port support requires server IP address on a default VRF.

Once a storage volume is enabled, applications can use the volume to store retrieve and delete files and directories.

External storage commands

address

Syntax

```
address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
no address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
```

Description

Specifies the NAS IP address or hostname.

The `no` form of this command deletes an IP address or hostname.

Command context

`config-external-storage-<VOLUME-NAME>`

Parameters

<IPV4-ADDR>

Specifies the NAS server IPv4 address, Global.

<IPV6-ADDR>

Specifies the IPv6 address of the NAS server.

<HOSTNAME>

Specifies the hostname of the NAS server. String.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating the logfiles storage volume with IP address 10.1.1.1:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# address 10.1.1.1
```

Deleting an external storage volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no address 10.1.1.1
```

directory

Syntax

```
directory <DIRECTORY-NAME>
no directory <DIRECTORY-NAME>
```

Description

Selects an existing directory on the external storage volume.

The `no` form of this command clears a directory of an external storage volume.

Command context

`config-external-storage-<VOLUME-NAME>`

Parameters

<DIRECTORY-NAME>

Specifies the external storage directory for mapping the volume.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Creating a volume named logfiles that is mapped under /home on the server:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# directory /home
```

Clearing the directory /home:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no directory /home
```

disable

Syntax

```
disable
no disable
```

Description

Disables the external storage volume.

The `no` form of this command enables the external storage volume. This is identical to the `enable` command.

Command context

`config-external-storage-<VOLUME-NAME>`

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Disabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

enable

Syntax

```
enable
no enable
```

Description

Enables the external storage volume.

The `no` form of this command disables the external storage volume. This is identical to the `disable` command.

Command context

`config-external-storage-<VOLUME-NAME>`

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Creating and then enabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# enable
```

Disables the external storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

external-storage

Syntax

```
external-storage <VOLUME-NAME>
no external-storage <VOLUME-NAME>
```

Description

Creates or updates an external storage volume.

The `no` form of this command deletes an external storage volume.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating the logfiles storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)#
```

Deleting the logfiles storage volume:

```
switch(config)# no external-storage logfiles
```

password

Syntax

```
password [{plaintext | ciphertext} <PASSWORD>]
no password
```

Description

Sets the password for logging in to a network attached storage server.

The `no` form of this command clears the password for logging in to a network attached storage server.

Command context

`config-external-storage-<VOLUME-NAME>`

Parameters

`{ciphertext | plaintext}`

Selects the password format.

`<PASSWORD>`

Specifies the password.

When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a volume named logfiles with password Xj#9:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# password plaintext Xj#9
```

Creating a volume named bak1 with a prompted plaintext password:

```
switch(config)# external-storage bak1
switch(config-external-storage-logfiles)# password
Enter the NAS server password: **********
Re-Enter Enter the NAS server password: **********
```

Clearing the password for volume logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no password
```

show external-storage

Syntax

`show external-storage [<VOLUME-NAME>]`

Description

Shows external storage configuration and state for all volumes or for a specified volume.

Command context

Operator (>) or Manager (#)

Parameters

<VOLUME-NAME>

Specifies the external storage volume name that the show command will use.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# show external-storage

--------------------------------------------------------------------------------
```

```
            Address       VRF      Username       Type        Directory      State
-------------------------------------------------------------------------------------
nfsvol      10.1.1.1      nas      ---            NFSv3       /home          operational
nfsfiles    20.1.1.1      nas      netstorage     NFSv4       /netstor       disabled
scpdev      nasserver     nas      scpstor        SCP         /scp           unaccessible
```

show running-config external-storage

Syntax

`show running-config external-storage`

Description

Shows the running configuration of the external storage.

Command context

Operator (`>`) or Manager (`#`)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# show running-config external-storage

external-storage nfsvol
      address   10.1.1.1
      vrf       nas
      type      nfsv4
      directoty /home
      enable
external-storage scpdev
      address   30.1.1.1
      vrf       nas
      username  switchuser
      password  ciphertext xxx
      type      scp
      directoty /home
      enable
```

type

Syntax

```
type {nfsv3 | nfsv4 | scp}
no type {nfsv3 | nfsv4 | scp}
```

Description

Sets the network attached storage access type for reaching the external storage volume.

The `no` form of this command deletes an external storage volume.

Command context

`config-external-storage-<VOLUME-NAME>`

Parameters

nfsv3

Specifies the NFSv3 network access protocol.

nfsv4

Specifies the NFSv4 network access protocol.

scp

Specifies the SCP network access protocol.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating the logfiles volume using NFSV4:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# type nfsv4
```

Clearing the external storage access type:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no type nfsv4
```

username

Syntax

```
username <USER-NAME>
no username <USER-NAME>
```

Description

Sets the username for logging in to a network attached storage server.

The `no` form of this command clears a username.

Command context

`config-external-storage-<VOLUME-NAME>`

Parameters

<USER-NAME>

Specifies the username.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a volume named logfiles with the user name nassuser:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# username nasuser
```

Clearing the user name nasuser from accessing the logfiles volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no username nasuser
```

vrf

Syntax

```
vrf <VRF-NAME>
no vrf <VRF-NAME>
```

Description

Setting a VRF to reach network attached storage.

The `no` form of this command clears access of a VRF to network attached storage.

Command context

`config-external-storage-<VOLUME-NAME>`

Parameters

<VRF-NAME>

Specifies the VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating the logfiles volume and setting a VRF named nas to access the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# vrf nas
```

Clearing access of a VRF named nas to the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no vrf nas
```

Chapter 7 IP-SLA

The IP Service Level Agreement (IP-SLA) is a feature that enables the measuring of network performance between two nodes in a network for different service level agreement parameters such as round-trip time (RTT), one-way delay, jitter, reachability, packet loss, and voice quality scores. These two nodes can span across area in access, distribution or core inside a LAN as well as across WAN between core to core or core to Data Centre switches. This feature helps you measure the SLA for different protocols or applications such as UDP echo, UDP jitter (for voice and video), TCP connect, HTTP, and ICMP echo. This guide provides details for managing and monitoring different types of IP-SLAs.

IP-SLA guidelines

- ArubaOS-CX supports only SLA configuration through CLI and thresholds can be configured using NAE agents using WebUI/REST.
- ArubaOS-CX supports only forever tests. On-demand tests are not supported.
- Maximum sessions: IP-SLA source 500, IP-SLA responder 100.
- NAE can effectively monitor a maximum of 300 parameters, reducing the maximum supported session by 300.
- NAE supports only syslog.
- NAE agents must be triggered for each IP-SLA test on every switch.
- If multiple IP addresses are received for a DNS query, DNS works with the first resolved IP.
- When the DNS server IP is not configured, the first DNS server in `resolve.conf` is used.
- The source interface/IP option is not applicable for SLAs configured on 'mgmt' VRF, as it has only one interface.
- A system time change because of NTP or a manual change causes an incorrect calculation.
- There is no interoperability of UDP echo SLA between ArubaOS-CX and FlexFabric switches.
- Source IP and source port combination must be unique across SLA sessions in a same switch.
- Do not use the same source port across the source and responder sessions in a switch.
- NTP synchronization is a must for SLA types involving one-way delay such as UDP jitter VoIP.
- It is mandatory to set default CoPP to the max value when UDP jitter SLA is enabled otherwise 100% packet loss can be seen and `UDP-Jitter sla` probe will result in failure as seen in the following example.

```
copp-policy default
     class hypertext priority 6 rate 50000 burst 64
     default-class priority 6 rate 99999 burst 9999
```

- Deviations with respect to PVOS results: The packet losses due to internal switch-related issues like interface shutdown or interface flaps will not be considered as 'Probes Timed-out error', as the IP-SLA solution is to measure network performance and anomalies. Rather, this kind of packet loss will be counted in internal counters like 'Destination address unreachable'.

Limitations with VoIP SLAs

- A maximum of 80 concurrent VoIP SLAs can be scheduled in a 20 second slot.
- A single VoIP probe takes 20 seconds to complete.
- The default and minimum probe interval for VoIP SLA is 120 seconds.
- SLAs scheduled in the same slot, periodically sends 1000 probe packets for 120 seconds in 20 second intervals.
- Default 120 second probe interval is divided in to 6 slots of 20 seconds to avoid synchronization of all configured VoIP SLAs sending probes at the same time.
- SLAs started at the same time exceeding the concurrent limit of 80 must wait for the next 20 second VoIP slot to open before moving to ‘running’ state.
- The maximum number of VoIP SLAs supported is 80 X 6 slots = 480 SLAs.
- SLAs exceeding 480 will continue to remain in the 'waiting for VoIP slot' until any slot is freed by stopping the running SLA.
- To avoid high RTT, a single switch with more than 20 SLAs should not have single responder SLA.
- When IP is received dynamically (e.g. using DHCP) for interfaces other than management interface, IPSLA source or responder has to be configured only using interface name.

IP-SLA commands

http

Syntax

```
http {get | raw} URL [source {<SOURCE-IPV4-ADDR> | <IFNAME>} source-port <PORT-NUM>]
     [proxy proxy-url] [cache disable] [name-server <IPV4-ADDR-DNS-SERVER>]
     [probe-interval <30-604800>] [version<VERSION-NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

Description

Configures HTTP as the IP-SLA test mechanism. Requires destination URL and type of HTTP request (raw/get).

Command context

`config-ip-sla-<IP-SLA-NAME>`

Parameters

`{get | raw}`

Selects HTTP request type as GET or RAW where the system will generate or provide HTTP payload.

`URL`

Specifies HTTP URL address of syntax. http://<HOST NAME/IP-ADDRESS>:<PORT>/<PATH>.

source `{<SOURCE-IPV4-ADDR> | <IFNAME>}`

Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.

source-port `<PORT-NUM>`

Specifies the value of the source port for the IP-SLA probes.

`cache disable`

Selects cache option for the HTTP server. By default the option is enabled.

name-server `<IPV4-ADDR-DNS-SERVER>`

Specifies the IPv4 address of DNS server.

probe-interval *<PROBE-INTERVAL>*

Specifies the probe interval in seconds. Range: 30 to 604800.

version *<VERSION-NUMBER>*

Specifies the source interface to use for sending IP-SLA probes.

http-raw-request *<RAW-PAYLOAD>*

HTTP raw request. String.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-ipsla-1)# http get http://device.arubanetworks.com/root/home.html
    switch(config-ipsla-1)# http raw http://device.arubanetworks.com/root/home.html
    switch(config-ipsla-1)# http 2.2.2.2 source 1/1/1
    switch(config-ipsla-1)# http http://device.arubanetworks.com  source 2.2.2.1
    switch(config-ipsla-1)# http http://device.arubanetworks.com/root/home.html
source-interface 1/1/1
    switch(config-ipsla-1)# http http://device.arubanetworks.com  name-server
10.10.10.2
    switch(config-ipsla-1)# http raw raw-request "GET /en/US/hmpgs/index.html
HTTP/1.0\r\n\r\n"
```

icmp-echo

Syntax

```
icmp-echo {<DEST-IPV4-ADDR>|<HOSTNAME>} [source {<SOURCE-IPV4-ADDR> | <IFNAME>}]
     [name-server <IPV4-ADDR-DNS-SERVER>] [payload-size <PAYLOAD-SIZE>]
     [tos <TYPE-OF-SERVICE>] [probe-interval <PROBE-INTERVAL>]
```

Description

Configures ICMP echo as the IP-SLA test mechanism. Requires destination address for the IP-SLA test.

Command context

`config-ip-sla-<IP-SLA-NAME>`

Parameters

{*<DEST-IPV4-ADDR>* **|** *<HOSTNAME>***}**

Selects the destination IPv4 address for the IP-SLA or the hostname of the destination.

[source {*<SOURCE-IPV4-ADDR>* **|** *<IFNAME>***}]**

Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.

name-server *<IPV4-ADDR-DNS-SERVER>*

Specifies the DNS server for destination hostname resolution.

payload-size *<PAYLOAD-SIZE>*

Specifies the payload size of an SLA probe. Range: 0 to 1440.

tos *<TYPE-OF-SERVICE>*

Specifies the type of serve to be used in the probe packets. Range: 0 to 255.

probe-interval *<PROBE-INTERVAL>*

Specifies the probe interval in seconds. Range: 5 to 604800.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ip-sla test
    switch(config-ip-sla-test)# icmp-echo 2.2.2.2
    switch(config-ip-sla-test)# icmp-echo 2.2.2.2 source 3.3.3.3
    switch(config-ip-sla-test)# icmp-echo 2.2.2.2 source 3.3.3.3 payload-size 400
    switch(config-ip-sla-test)# icmp-echo 2.2.2.2 source 3.3.3.3 payload-size 400
name-server 4.4.4.4
    switch(config-ip-sla-test)# icmp-echo 2.2.2.2 source 3.3.3.3 payload-size 400
name-server 4.4.4.4  probe-interval 80
```

ip-sla

Syntax

```
ip-sla <IP-SLA-NAME>
no ip-sla <IP-SLA-NAME>
```

Description

Creates an IP Service Level Agreement (SLA) profile and switches to the `config-ip-sla` context.

The `no` form of this command deletes an IP-SLA profile. By default, all profile use the default VRF (default).

Command context

`config`

Parameters

<IP-SLA-NAME>

Specifies an IP-SLA profile name. Length: 1 to 63 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an IP-SLA:

```
switch(config)# ip-sla 1
switch(config-ip-sla-1)#
```

Deleting an IP-SLA:

```
switch(config)# no ip-sla 1
switch(config)#
```

ip-sla responder

Syntax

```
ip-sla responder <SLA-NAME> {udp-echo | tcp-connect | udp-jitter-voip} <PORT-NUM>
     [source {<SOURCE-IPV4-ADDR> | <IFNAME>}][vrf <VRF-NAME>]
no ip-sla responder <SLA-NAME>
```

Description

Selects the IP-SLA responder. The responder can be configured for udp-echo, tcp-connect, udp-jitter-voip type. It requires the SLA name, SLA type, and port number as arguments. Source IP/interface ID is a must for type udp-jitter-voip and optional for other types.

The `no` form of this command removes the IP-SLA responder.

Command context

`config`

Parameters

<SLA-NAME>

Specifies the SLA name.

`udp-echo`

Enables responder for udp-echo probes.

`tcp-connect`

Selects TCP connect as the IP-SLA test mechanism.

`vrf` *<VRF-NAME>*

Specifies the name of the VRF to use.

`udp-jitter-voip`

Selects VOIP jitter as the IP-SLA test mechanism.

<PORT-NUM>

Specifies the port number to listen for IP-SLA probes. Range: 1 to 65535.

`[source {<SOURCE-IPV4-ADDR> | <IFNAME>}]`

Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 1/1/1
```

```
switch(config)# no ip-sla responder <SLA-NAME>
```

show ip-sla responder

Syntax

`show ip-sla responder <SLA-NAME>`

Description

Shows the given IP-SLA responder configuration and operation status.

Command context

`config`

Parameters

<SLA-NAME>

Specifies the SLA name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# show ip-sla responder SLA3

    SLA Name             : SLA3
    IP-SLA Type          : Udp-echo
    VRF                  : Default
    Responder Port       : 8000
    Responder IP         : 2.2.2.3
    Responder Interface  : 1/1/1
    Responder Status     : Running
```

show ip-sla responder results

Syntax

`show ip-sla responder <SLA-NAME> <SOURCE-IPV4-ADDR> <PORT-NUM> results`

Description

Shows the given ip-sla responder statistics for a given source IP and port. This command is only applicable for the sources where source IP and port are configured.

Command context

`config`

Parameters

<SLA-NAME>

Specifies the SLA name.

<SOURCE-IPV4-ADDR>

Specifies the source IPV4 address.

<PORT-NUM>

Specifies the port number. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# show  ip-sla responder SLA1 2.2.2.1 8000 results

    IP-SLA Type          : Udp-echo
    VRF Name             : Default
    Source IP            : 2.2.2.1
    Source Port          : 8000
    Responder Port       : 8888
    Responder IP         : 2.2.2.3
    Responder Interface  :
    Responder Status     : Running
    Packets Received     : 2
    Packets Sent         : 2
```

show ip-sla <*SLA-NAME*>

Syntax

```
show ip-sla <SLA-NAME> results
```

Description

Shows the given IP-SLA source configuration and status.

Command context

Operator (>) or Manager (#)

Parameters

<SLA-NAME>

Specifies the SLA name.

`results`

Shows the statistics calculated for an SLA type.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# show ip-sla xyz results

    IP-SLA session status
       IP-SLA Name                     : xyz
       IP-SLA Type                     : tcp-connect
       Destination Host Name/IP Address: 2.2.2.1
       Destination Port                : 8888
       Source IP Address/IFName        : 2.2.2.2
       Source Port                     : 5555
       Status                          : Running

    IP-SLA session cumulative counters
       Total Probes Transmitted        : 1
       Probes Timed-out                : 0
       Bind Error                      : 0
       Destination Address Unreachable : 0
       DNS Resolution Failures         : 0
       Reception Error                 : 0
       Transmission Error              : 0

    IP-SLA Latest Probe Results
       Last Probe Time                 : 2018 Jul 13 02:00:35
       Packets Sent                    : 1
       Packets Received                : 1
       Packet Loss in Test             : 0.0000%

    Minimum RTT(ms)                    : 0.7900
    Maximum RTT(ms)                    : 0.7900
    Average RTT(ms)                    : 0.7900
    DNS RTT(ms)                        : 0.0000
    TCP RTT(ms)                        : 0.9710

switch(config)# show ip-sla xyz
    IP-SLA Name             : xyz
```

```
    Status                  : scheduled
    IP-SLA Type             : tcp-connect
    VRF                     : ipslasrc
    Source Port             : 5555
    Source IP               : 2.2.2.2
    Source Interface        :
    Domain Name Server      :
    Probe interval(seconds) : 90


switch(config)# show ip-sla jitter-sla results
    IP-SLA session status
       IP-SLA Name                        : jitter-sla
       IP-SLA Type                        : udp-jitter-voip
       Destination Host Name/IP Address: 2.2.2.1
       Destination Port                   : 8888
       Source IP Address/IFName           :
       Source Port                        : 5555
       Status                             : Running

    IP-SLA Session Cumulative Counters
       Total Probes Transmitted           : 1
       Probes Timed-out                   : 0
       Bind Error                         : 0
       Destination Address Unreachable : 0
       DNS Resolution Failures            : 0
       Reception Error                    : 0
       Transmission Error                 : 0

    IP-SLA Latest Probe Results
       Last Probe Time                    : 2018 Jul 13 02:02:48
       Packets Sent                       : 1
       Packets Received                   : 1
       Packet Loss in Test                : 0.0000%

       Minimum RTT(ms)                    : 0.7900
       Maximum RTT(ms)                    : 0.7900
       Average RTT(ms)                    : 0.7900
       DNS RTT(ms)                        : 0.0000

       Min Positive SD                    : 1      Min Positive DS        : 2
       Max Positive SD                    : 1      Max Positive DS        : 2
       Positive SD Number                 : 2      Positive DS Number     : 2
       Positive SD Sum                    : 2      Positive DS Sum        : 4
       Positive SD Average                : 5      Positive DS Average    : 5
       Min Negative SD                    : 1      Min Negative DS        : 1
       Max Negative SD                    : 1      Max Negative DS        : 1
       Negative SD Number                 : 2      Negative DS Number     : 4
       Negative SD Sum                    : 2      Negative DS Sum        : 4
       Negative SD Average                : 5      Negative DS Average    : 5

       Max SD Delay                       : 0      Max DS Delay           : 0
       Min SD Delay                       : 0      Min DS Delay           : 0
       Average SD Delay                   : 0      Average DS Delay       : 0

    Voice Scores:
       MOS  Score                         : 4.38   ICPIF                  : 0


switch(config)# show ip-sla m3op
    IP-SLA Name             : jitter-sla
    Status                  : Running
```

```
    IP-SLA Type              : udp-jitter-voip
    VRF                      : ipslasrc
    Source IP                : 2.2.2.2
    Source Interface         :
    Domain Name Server       :
    TOS                      : 10
    Probe Interval(seconds)  : 90
    Advantage Factor         : 0
    Codec Type               : g711a


switch(config)# show ip-sla http-sla
    IP-SLA Name              : http-sla
    Status                   : Running
    IP-SLA Type              : http
    VRF                      : ipslasrc
    Source IP                : 2.2.2.2
    Source Interface         :
    Domain Name Server       : 10.10.10.2
    Probe Interval(seconds)  : 90
    HTTP Request Type        : GET
    HTTP/HTTPS URL           : abcd.com/ws/home
    Cache                    : Enabled
    HTTP Proxy URL           :
    HTTP Version Number      : 1.1
    ```

IP-SLA status description
    ```
    | Status                  | Description                                      |
    |-------------------------|--------------------------------------------------|
    | Running                 | SLA is fully operational                         |
    | Bind Error              | Another service is using the same source port   |
    | Interface Down          | Interface status is not up
    | Dns Resolution Error    | Failed to resolve destination hostname          |
    | No Route                | No available route to the responder             |
    | Internal Error          | Unexpected error prevents SLA session           |
    | Disabled                | SLA is disabled                                  |
    |Configuration Incomplete | Configuration is not complete to enable the SLA|
    ```
IP SLA session cumulative counters description
    ```
    | Status                         | Description
                             |
    |--------------------------------|----------------------------------------------
----------------------------|
    |Probes Timed-out                | Total numbers of probes failed to receive
response.                       |
    |Bind Error                      | Total numbers of probes transmission failed
as source port not available.|
    |Destination Address Unreachable | Total numbers of probes transmission failed
due to route unavailable.    |
    |DNS Resolution Failures         | Total numbers of probes failed due to DNS
resolution failure.             |
    |Reception Error                 | Total numbers of probes failed due to
internal error in reception.       |
    |Transmission Error              | Total numbers of probes failed due to
internal errr in transmission.     |
```

start-test

Syntax

start-test

Description

Starts the IP-SLA probes.

Command context

config-ip-sla-*<IP-SLA-NAME>*

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# start-test
```

stop-test

Syntax

stop-test

Description

Stops the IP-SLA probes.

Command context

config-ip-sla-*<IP-SLA-NAME>*

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# stop-test
```

tcp-connect

Syntax

```
tcp-connect {<DEST-IPV4-ADDR> | <HOSTNAME>} <PORT-NUM> [source {<SOURCE-IPV4-ADDR> |
     <IFNAME>} [source-port <PORT-NUM>]] [name-server <IPV4-ADDR-DNS-SERVER>]
     [probe-interval <PROBE-INTERVAL>]
```

Description

Configures TCP connect as the IP-SLA test mechanism. Requires destination address/hostname and destination port for the IP-SLA of tcp-connect IP-SLA type.

Command context

config-ip-sla-*<IP-SLA-NAME>*

Parameters

{*<DEST-IPV4-ADDR>* **|** *<HOSTNAME>***}**

Selects the destination IPv4 address for the IP-SLA or the hostname of the destination.

<PORT-NUM>

Destination port for the IP-SLA. Range: 1 to 65535.

[source {*<SOURCE-IPV4-ADDR>* | *<IFNAME>*}]

Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.

[source-port *<PORT-NUM>*]

Specifies the port for the IP-SLA test.

[name-server *<IPV4-ADDR-DNS-SERVER>*]

Specifies the DNS server for destination hostname resolution.

[probe-interval *<PROBE-INTERVAL>*]

Probe interval in seconds. Range: 30 to 604800.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-ipsla-1)# tcp-connect 2.2.2.2 8080
       switch(config-ipsla-1)# tcp-connect 2.2.2.2 8080 source 2.2.2.1 source-port
6000
       switch(config-ipsla-1)# tcp-connect 2.2.2.2 8080 source 1/1/1 source-port
6000

       switch(config-ipsla-1)# tcp-connect https://device.arubanetworks.com 8080
       switch(config-ipsla-1)# tcp-connect https://device.arubanetworks.com 8080
source 2.2.2.1 source-port 6000
       switch(config-ipsla-1)# tcp-connect https://device.arubanetworks.com 8080
source 1/1/1 source-port 6000
       switch(config-ipsla-1)# tcp-connect https://device.arubanetworks.com 8080
name-server 10.10.10.2
```

udp-echo

Syntax

```
udp-echo {<DEST-IPV4-ADDR>|<HOSTNAME>} <PORT-NUM> [source {<SOURCE-IPV4-ADDR> |
     <IFNAME>} [source-port <PORT-NUM>]] [name-server <IPV4-ADDR-DNS-SERVER>] [payload-size
     <PAYLOAD-SIZE>] [tos <TYPE-OF-SERVICE>] [probe-interval <PROBE-INTERVAL>]
```

Description

Configures UDP echo as the IP-SLA test mechanism. Requires destination address/hostname and destination port number for the IP-SLA of udp-echo SLA type.

Command context

`config-ip-sla-<IP-SLA-NAME>`

Parameters

{*<DEST-IPV4-ADDR>* | *<HOSTNAME>*}

Selects the destination IPv4 address for the IP-SLA or the hostname of the destination.

<PORT-NUM>

Specifies the destination port for the IP-SLA. Range: 1 to 65535.

[source {*<SOURCE-IPV4-ADDR>* | *<IFNAME>*}]

Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.

[source-port *<PORT-NUM>*]

Specifies source port for the IP-SLA test. Range: 1 to 65535.

`[name-server <IPV4-ADDR-DNS-SERVER>]`

Specifies the DNS server for destination hostname resolution.

`[payload-size <PAYLOAD-SIZE>]`

Specifies the payload size of an SLA probe. Range: 28 to 1440.

`[<TYPE-OF-SERVICE>]`

Type of service. Range: 0 to 255.

`probe-interval <PROBE-INTERVAL>`

Probe interval in seconds. Range: 5 to 604800.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-ipsla-1)# udp-echo 2.2.2.2 8080
    switch(config-ipsla-1)# udp-echo 2.2.2.2 8080 source 2.2.2.1
    switch(config-ipsla-1)# udp-echo https://device.arubanetworks.com 8080
    switch(config-ipsla-1)# udp-echo 2.2.2.2 8080 source 1/1/1
    switch(config-ipsla-1)# udp-echo 2.2.2.2 8080 source 2.2.2.1 payload-size 50
    switch(config-ipsla-1)# udp-echo 2.2.2.2 8080 source 1/1/1 payload-size 50
    switch(config-ipsla-1)# udp-echo 2.2.2.2 8080 payload-size 50
    switch(config-ipsla-1)# udp-echo https://device.arubanetworks.com 8080 source
2.2.2.1
     payload-size 50
    switch(config-ipsla-1)# udp-echo https://device.arubanetworks.com 8080 source
1/1/1
     payload-size 50
    switch(config-ipsla-1)# udp-echo https://device.arubanetworks.com 8080
     name-server 10.10.10.2
```

udp-jitter-voip

Syntax

```
udp-jitter-voip {<DEST-IPV4-ADDR> | <HOSTNAME>} <PORT-NUM> [codec-type <CODEC-TYPE>]
     [advantage-factor <VALUE>] [source {<SOURCE-IPV4-ADDR> | <IFNAME>} [source-port <PORT-
NUM>]]
     [name-server <IPV4-ADDR-DNS-SERVER>][probe-interval <PROBE-INTERVAL>] [tos <TYPE-OF-
SERVICE>]
```

Description

Configure UDP jitter voip as the IP-SLA test mechanism. Requires destination address/hostname and source address/interface for the IP-SLA of udp-jitter-voip IP-SLA type.

Command context

`config-ip-sla-<IP-SLA-NAME>`

Parameters

`{<DEST-IPV4-ADDR>|<HOSTNAME>}`

Selects the destination IPv4 address for the IP-SLA or the hostname of the destination.

`<PORT-NUM>`

Selects the port number for the IP-SLA. Range: 1 to 65535.

`[codec-type <CODEC-TYPE>]`

Selects the codec-type for the Voip IP-SLA test.

`[advantage-factor <ADVANTAGE-FACTOR>]`

Selects the value for the advantage factor. Default value is 0.

`[source {<SOURCE-IPV4-ADDR> | <IFNAME>}]`

Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.

`[source-port <PORT-NUM>]`

Specifies the value of source port for the IP-SLA probes.

`[name-server <IPV4-ADDR-DNS-SERVER>]`

Specifies the DNS server for destination hostname resolution.

`tos <TYPE-OF-SERVICE>`

Specifies the type of service. Range: 0 to 255.

`probe-interval <PROBE-INTERVAL>`

Specifies the probe interval in seconds. Range: 120 to 604800.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-ipsla-1)# udp-jitter-voip  2.2.2.2 8080 advantage-factor 10 codec-type
g711a
    switch(config-ipsla-1)# udp-jitter-voip  2.2.2.2 8080 advantage-factor 10 codec-
type g711a source 2.2.2.1
    switch(config-ipsla-1)# udp-jitter-voip  https://device.arubanetworks.com 8080
advantage-factor 10 codec-type g711a
    switch(config-ipsla-1)# udp-jitter-voip  2.2.2.2 8080 advantage-factor 10 codec-
type g711a source 1/1/1
    switch(config-ipsla-1)# udp-jitter-voip  https://device.arubanetworks.com 8080
advantage-factor 10 codec-type g711a source 2.2.2.1
    switch(config-ipsla-1)# udp-jitter-voip  https://device.arubanetworks.com 8080
advantage-factor 10 codec-type g711a source 1/1/1
    switch(config-ipsla-1)# udp-jitter-voip  https://device.arubanetworks.com 8080
advantage-factor 10 codec-type g711a name-server 10.10.10.2 probe-interval 120
source 10.1.1.1 source-port 8888 tos 10
```

vrf

Syntax

```
vrf <VRF-NAME>
no vrf [<VRF-NAME>]
```

Description

Configures the VRF on which the SLA will send or receive packets. By default, the default VRF is used.

The `no` form of the command removes VRF from SLA.

Command context

`config-ip-sla-<IP-SLA-NAME>`

Parameters

`<VRF-NAME>`

Specifies a VRF name. Length: Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-ip-sla-test)# vrf ipslasrc
```

```
switch(config-ip-sla-test)# no vrf
```

Chapter 8
L1-100Mbps downshift

The speed downshift feature allows the user to link-up at sub-optimal speeds when failing to link-up at the highest advertised speed. There are fixed number of link attempts made to establish link at highest advertised speed and when all of them fail and attempt is made to link-up at a lower possible speed.

This feature requires underlying PHY to have support for the same and hence capability is only added to select set of ports. If a link cannot be established at the highest common denominator within a set number of link attempts, the PHY advertises the next highest speed using auto-negotiation.

Limitations with speed downshift

- Link up may be delayed as certain number of retries are done to establish the link at highest advertise speeds by both link partners before downshifting.
- Link may be established at sub-optimal speed.

L1-100Mbps downshift commands

downshift enable

Syntax

```
downshift-enable
no downshift-enable
```

Description

Enables/disables automatic speed downshift on an interface that supports downshift, generally 1GBASE-T ports. When enabled, downshift allows an interface to link at a lower advertised speed when unable to establish a stable link at the maximum speed. Downshifting only applies to physical interfaces that are not members of a LAG and is only available when auto-negotiation is enabled. When only one speed is advertised, downshift will not be triggered.

Command context

```
config-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-if)# interface 1/1/1
switch(config-if)# downshift-enable

Warning: this is a non-standard mode for use only when standards-based
auto-negotiation is not able to establish a stable link. Enabling this
```

```
may cause the port to link at a lower than expected speed and should
not be used on ports that are members of a LAG. Support calls may require
this feature to be disabled

Continue (y/n)?

switch(config-if)#
```

When automatic downshift is enabled:

```
switch(config-if)# show running-config interface
interface 1/1/1
    downshift-enable
```

Disabling automatic speed downshift:

```
switch(config-if)# interface 1/1/1
switch(config-if)# no downshift-enable
```

show interface

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical | extended [non-zero]]
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief | physical]
show interface [lag | loopback | tunnel | vlan ] [<ID>] [extended | non-zero]
```

Description

Displays active configurations and operational status information for interfaces.

Command context

`config`

Parameters

<IFNAME>

Specifies a interface name.

<IFRANGE>

Specifies the port identifier range.

`brief`

Displays brief info in tabular format.

`extended`

Displays the physical connection info in tabular format.

`non-zero`

Displays only non zero statistics.

`LAG`

Displays LAG interface information.

`LOOPBACK`

Displays loopback interface information.

`TUNNEL`

Displays tunnel interface information.

`VLAN`

Displays VLAN interface information.

<LAG-ID>

Specifies the LAG number. Range: 1-256

<LOOPBACK-ID>

Specifies the LOOPBACK number. Range: 0-255

<TUNNEL-ID>

Specifies the tunnel ID. Range: 1-255

<VLAN-ID>

Specifies the VLAN ID. Range: 1-4094

`VXLAN`

Displays the VXLAN interface information.

<VXLAN-ID>

Specifies the VXLAN interface identifier. Default: 1

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing interfaces configured for L2 forwarding:

```
switch(config-if)# show interface
Interface 1/1/1 is down (Administratively down)
 Admin state is down
 State information: Administratively down
 Link transitions: 2
 Description: Backup data center link
 Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
 MTU 9198
 Type SFP-SX
 Full-duplex
 qos trust none
 Speed 0 Mb/s
 Auto-negotiation is on
 Flow-control: off
 Error-control: off
 VLAN Mode: access
 Access VLAN: 1
 Rx
      1386055 total packets            586397526 total bytes
      1374287   unicast packets
        11764   multicast packets
            4   broadcast packets
            0 errors                           0 dropped
            0 CRC/FCS
 Tx
      2475612 total packets           1003506711 total bytes
      2311806   unicast packets
       100369   multicast packets
        63437   broadcast packets
            0 errors                     2462773 dropped
            0 collision
```

```
Interface 1/2/1 is down (Administratively down)
 Admin state is down
 State information: Administratively down
 Link transitions: 0
 Description:
 Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
 MTU 9198
 Type QSFP+SR4
 Full-duplex
 qos trust none
 Speed 0 Mb/s
 Auto-negotiation is off
 Flow-control: off
 Error-control: off
 VLAN Mode: access
 Access VLAN: 1
 Rx
           0 total packets                   0 total bytes
           0   unicast packets
           0   multicast packets
           0   broadcast packets
           0 errors                          0 dropped
           0 CRC/FCS
 Tx
           0 total packets                   0 total bytes
           0   unicast packets
           0   multicast packets
           0   broadcast packets
           0 errors                          0 dropped
           0 collision
```

When the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1

Interface 1/1/1 is up
 ...
 Auto-negotiation is on with downshift active
```

show interface downshift-enable

Syntax

`show interface [<IFNNAME>|<IFRANGE>] downshift-enable`

Description

Displays speed downshift information, including the interface speed status and configuration.

Command context

`config`

Parameters

<IFNAME>

Specifies a interface name.

<IFRANGE>

Specifies the port identifier range.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing automatic downshift information:

```
switch(config-if)# show interface downshift-enable
-------------------------------------------------
              Downshift               Speed
Port      Enabled | Active    Status    | Config
-------------------------------------------------
1/1/1     yes       yes       100M-FDx   auto
1/1/2     yes       no        1G         auto
1/1/3     yes       no        100M-FDx   100M-FDx
1/1/4     no        no        --         auto
```

Showing automatic downshift information on per interface:

```
switch(config-if)# show interface 1/1/2 downshift-enable
-------------------------------------------------
              Downshift               Speed
Port      Enabled | Active    Status    | Config
-------------------------------------------------
1/1/2     yes       no        1G         auto
```

show running-config interface

Syntax

```
show running-config interface [<IFNNAME>|<IFRANGE>]
show running-config interface [lag | loopback | tunnel | vlan ] [<ID>]
```

Description

Displays active configurations of various switch interfaces.

Command context

`config`

Parameters

<IFNAME>

Specifies a interface name.

<IFRANGE>

Specifies the port identifier range.

`LAG`

Specifies LAG interface information

`LOOPBACK`

Specifies loopback interface information.

`TUNNEL`

Specifies tunnel interface information.

`VLAN`

Specifies VLAN interface information.

<LAG-ID>

Specifies the LAG number. Range: 1-256.

<LOOPBACK-ID>

Specifies the LOOPBACK number. Range: 0-255.

<TUNNEL-ID>

Specifies the tunnel ID. Range: 1-255.

<VLAN-ID>

Specifies the VLAN ID. Range: 1-4094.

VXLAN

Specifies the VXLAN interface information.

<VXLAN-ID>

Specifies the VXLAN interface identifier. Default: 1.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing 1/1/2 interface configuration:

```
switch(config-if)# show running-config interface 1/1/2

interface 1/1/2
   no shutdown
   description DC-23
   exit
```

Showing loopback interfaces configured:

```
switch(config-if)# show running-config interface loopback

interface loopback 1
    description lb interface 1
    exit
interface loopback 2
    description lb interface 2
    exit
```

Showing loopback interfaces not configured:

```
switch(config-if)# show running-config interface loopback

No loopback interfaces configured.
```

Chapter 9 Mirroring

Mirroring allows you to replicate all traffic arriving and/or leaving the selected system interfaces. This data can be used for collection or analysis.

The traffic replicated using mirroring can be sent to a separate interface on the same switch as the traffic source for analysis or inspection. Such a collection of interfaces and settings is called a mirror session.

A mirror session can be configured with many traffic sources but only a single output, or destination. In the initial configuration, the mirror session is disabled. You have enable the feature to start the replication.

Care must be taken in choosing the number and rates of sources to avoid over-saturating a session destination. A mirror session with multiple 10G sources can overwhelm a single 10G destination and important data may be lost.

Mirror statistics

Mirror statistics are reset for a Mirror-to-CPU session when an interface is added or removed from a LAG that is a source interface in the Mirror session and during a failover.

Mirror endpoints

Enables switches to terminate a tunnel, decapsulate the original packet from the tunnel, and replicate the original packet out to one or more switch interface(s). Supports termination of GRE encapsulation.

Scenario 1

A topology where there are two switches (sw1,sw2) connected by one link.

sw1 contains the mirror configuration with tunnel as destination, the mirrored packets are GRE encapsulated.

sw2 contains the remote mirror termination, which decapsulates the packet and copies it to the configured destinations. In the below configuration, the remote mirror decapsulates the GRE packet with source/destination IP 9.9.9.9/2.2.2.2 with GRE Key 1 and replicate the inner payload to 1/1/2,1/1/3.

Configuration in sw1:

```
mirror session 1
destination tunnel 2.2.2.2 source 9.9.9.9
source interface 1/1/7 rx
enable
```

Configuration in sw2:

```
mirror endpoint test
source 9.9.9.9 destination 2.2.2.2 id 1
destination interface 1/1/2,1/1/3
```

```
no shutdown
```

Classifier policies and mirroring sessions

Network traffic can be mirrored to a destination interface in two ways:

- Using a mirroring session alone.
- Using Classifier Policies with mirror actions in conjunction with a mirroring session.

Basic mirroring sessions provide coarse control over the type of traffic mirrored from a source: all received, all transmitted, or both. However, a traffic class within a Classifier Policy applied to a source can provide much finer grained control of mirrored traffic. For example, a policy can match on many different aspects of the Ethernet or IPv4 or IPv6 header information in each frame or packet received or transmitted on an interface.

The steps to configure a policy and class with a mirror action are the following:

1. Configuring a mirroring session with a destination interface.
2. Enabling the mirroring session.
3. Configuring the Classifier Policy, specifying the mirroring session ID in the mirror action.

If the packets being mirrored are received from a VLAN that is not allowed on the mirror destination, the mirrored packets would be dropped at the mirror destination interface. When the mirrored packets are dropped at the destination, the mirror output packet and byte count will increment, however the packets will not be received at the mirror destination.

The mirror destination port among the active mirror sessions must be unique. That is, if an interface is configured as a source or destination in an active mirror session, the same port cannot be used as a destination in another active mirror session.

VLAN as a source

AOS-CX allows configuration of VLAN as a mirroring source. When a VLAN source is configured in the 'rx' direction, all packets are mirrored as they are received in the switch. When a VLAN source is configured in 'tx' direction, all packets are mirrored as they are transmitted out of the switch.

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 1024 source VLANs in each direction of a given mirror session.

Same VLANs can be configured as a mirror source for multiple sessions.

Note: When changing a source VLAN in an enabled mirror session (that is, adding, changing direction, or removing), mirrored packets being transmitted out the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways:

1. Reenter the `source vlan` command with the new preferred direction.
2. Use the `no` form of the command with a direction (rx or tx) to selectively remove the specified direction. Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

For packets bridged through the switch:

If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in the 'rx' direction, packets are mirrored in the pre-routed form with the destination MAC address as the switch address.
- If the mirror is configured in the 'tx' direction, packets are mirrored in the post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in the 'both' direction, one copy of the packet will be mirrored.

Control plane packets generated by the switch's CPU are processed both in the ingress and the egress packet processing pipeline. The following are the behaviors for mirroring with VLAN as source:

- If the mirror is configured in the 'rx' or 'tx' direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the 'both' direction, two copies of the packets are mirrored to the mirror destination.

Mirroring commands

clear mirror

Syntax

`clear mirror [all | <SESSION-ID>]`

Description

Clears the mirror statistics for all configured mirror sessions or a specified session

Command context

Manager (`#`)

Parameters

`all`

Specifies all configured sessions.

`<SESSION-ID>`

Specifies a numeric identifier for the session. Range: 1 to 4

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

clear mirror endpoint

Syntax

```
clear mirror endpoint [NAME]
```

Description

Clears mirror endpoint statistics for all configured mirror endpoints. The optional parameter can be added to clear a specific mirror endpoint.

Command context

Operator (>) or Manager (#)

Parameters

NAME

Specifies name of the mirror endpoint instance to be cleared.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing statistics for all configured mirror endpoints:

```
switch# clear mirror endpoint
```

Clearing mirror statistics for mirror endpoint test:

```
switch# clear mirror endpoint test
```

comment

Syntax

```
comment <COMMENT>
no comment
```

Description

Specifies a comment for the mirroring session.

When used in mirror endpoint command context, specifies a comment for the mirror endpoint.

The `no` form of this command removes the comment.

Command context

```
config-mirror-<SESSION-ID>
config-mirror-endpoint
```

Parameters

<COMMENT>

A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3)# comment This Mirror will be removed during next maintenance
window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3)# no comment
```

Adding a comment to a mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor endpoint traffic
```

Replacing the existing comment for mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor statistics on each endpoint
interfaces
```

copy tshark-pcap

Syntax

`copy tshark-pcap <REMOTE-URL> [vrf <VRF-NAME>]`

Description

Copies the tshark capture data to a file on a TFTP or SFTP server.

Command context

Manager (`#`)

Parameters

`<REMOTE-URL>`

Specifies the capture file on a remote TFTP or SFTP server. The URL syntax is:

{tftp:// | sftp://<*USER*>@} {<*IP*>|<*HOST*>} [:<*PORT*>] [;blocksize=<*SIZE*>]/<*FILE*>

`vrf` *`<VRF-NAME>`*

Specifies the name of a VRF. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Example

Copying the capture data to a file on SFTP server 10.0.0.2:

```
switch# copy tshark-pcap sftp://root@10.0.0.2/file.pcap

root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp> put packets.pcap file.pcap
Uploading packets.pcap to /root/file.pcap
packets.pcap                                  100%  156   219.8KB/s   00:00
Copied successfully.
```

destination cpu

Syntax

```
destination cpu
no destination cpu
```

Description

The command causes the mirror session to transmit mirrored packets to the switch CPU. This destination may be configured for multiple sessions, however only one such configured session may be active at a given time.

The diagnostic utility Tshark may be used to view and capture packets transmitted to the CPU through this route. Ctrl+C must be entered to terminate a Tshark capture session. More details can be found in the *Supportability Guide*.

The `no` form of this command will immediately stops mirroring traffic to the CPU, but will not remove any sources from the mirror configuration.

Command context

`config-mirror-<SESSION-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring a mirror session with CPU as the destination.

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination cpu
```

Removing the destination entirely.

```
switch(config-mirror-1)# no destination cpu
```

destination interface

Syntax

```
destination interface <IFNAMELIST>
no destination interface <IFNAMELIST>
```

<IFNAMELIST> can accept input as System interfaces and LAGs in the form of range, comma or both.

Description

config context:

Specifies the interface to where all mirrored traffic for the session will be transmitted.

There is a limit of 64 destination interfaces in a given mirror session.

You may configure the same destination in multiple mirror sessions however only one of those sessions sharing the destination can be enabled at a given time.

Layer 2 or Layer 3 Ethernet ports, LAGs, tunnels and CPU as a mirror destination is supported. A port that is already a member of a LAG is not a valid mirror destination.

Configuring a different destination interface in an enabled mirror session will cause all mirrored traffic to use the new interface. This may cause a temporary suspension of mirrored traffic from the sources during the reconfiguration.

The `no` form of this command will cease the use of the interface and disable the session. If there is at least one interface the mirror session will continue to be enabled.

mirror endpoint context:

Specifies the interface(s) to where the original packets after decapsulation from the tunnel will be transmitted. This includes system interfaces (e.g, 1/1/1) and LAG interfaces (e.g, lag1). A maximum of 64 interfaces can be configured as an endpoint interface for a given mirror endpoint.

The `no` form of this command remove the specific endpoint interface(s) from the configured list. If no interface is specified, all the endpoint interfaces will be removed.

Command context

`config-mirror-<SESSION-ID>`
`config-mirror-endpoint`

Parameters

`IFNAMELIST`

Specifies list of destination interfaces.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a mirror session and adding an interface as a destination:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with a different port:

```
switch(config-mirror-1)# destination interface 1/1/5
```

Replacing the existing destination with a LAG interface:

```
switch(config-mirror-1)# destination interface lag100
```

Removing the existing destination entirely:

```
switch(config-mirror-1)# no destination interface
```

Removing destination interfaces:

```
switch(config-mirror-1)# no destination interface 1/1/1-1/1/2
If an interface to delete is not part of the mirror session, display warning message
and continue

switch(config-mirror-1)# no destination interface 1/1/1-1/1/2
Ignoring destination interface 1/1/2 as it is not currently configured in mirror
session  1
```

Configuring 1/1/1-1/1/10,1/1/20 as a mirror endpoint interface:

```
switch(config-mirror-endpoint-test)# destination interface 1/1/1-1/1/10,1/1/20
```

Removing mirrored endpoint interface 1/1/1:

```
switch(config-mirror-endpoint-test)# no destination interface 1/1/1
```

Removing mirrored endpoint interfaces 1/1/1 -1/1/2:

```
switch(config-mirror-endpoint-test)# no destination interface 1/1/1-1/1/2
```

Attempting to remove 1/1/11 from a list when it is not configured:

```
switch(config-mirror-endpoint-test)# no destination interface 1/1/11
Ignoring interface 1/1/11 as it is not currently configured in mirror endpoint 1
```

diagnostic

Syntax

```
diagnostic

diag utilities tshark [file]
diag utilities tshark [delete-file]
```

Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.

The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the `delete-file` form of this command to delete the most recent capture file.

Since `file` and `delete-file` are optional, the behavior of the base command `diag utilities tshark` does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

Command context

Manager (`#`)

Parameters

`file`

Saves captured packets to a temporary file.

`delete-file`

Deletes the most recent captured file.

Authority

Administrators or local user group members with execution rights for this command.

Example

Performing diagnostic:

```
switch# diagnostic

switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

disable

Syntax

`disable`

Description

Disables the mirroring session specified by the current command context.

Command context

`config-mirror-<SESSION-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the `show mirror` command for that session ID shows an `Admin Status` of `disable` and an `Operation Status` of `disabled`.

Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

enable

Syntax

`enable`

Description

Enables the mirroring session for the current command context.

Command context

`config-mirror-<SESSION-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the `show mirror` command for that session ID shows an `Admin Status` of `enable` and an `Operation Status` of `enabled`.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of `rx` or `both`):

- The attempt to enable the mirroring session fails and an error is returned.

When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

Example

On the 6400 Switch Series, interface identification differs.

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# source interface 1/1/2 rx
switch(config-mirror-3)# destination interface 1/1/3
switch(config-mirror-3)# comment Monitor router port ingress-only traffic
switch(config-mirror-3)# enable
```

mirror session

Syntax

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The `no` form of this command removes an existing mirroring session from the configuration.

Command context

`config`

Parameters

<SESSION-ID>

Specifies the session identifier. Range: 1 to 4

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#

switch(config)# mirror session 3
switch(config-mirror-3)#

switch(config)# no mirror session 1
switch(config)#
```

mirror endpoint

Syntax

```
mirror endpoint NAME
no mirror endpoint NAME
```

Description

Creates the specified mirror endpoint or enters its context if it already exists. The specifics of a mirror endpoint are created or altered while in the mirror endpoint context and the mirror endpoint is enabled or disabled from this context. It may be possible to support different encapsulations by different ASICs. For example, UDP for PVOS compatibility. Termination of GRE encapsulation is also supported.

The `no` form of this command removes an existing mirror endpoint. An enabled mirror endpoint is automatically disabled first before removal.

Command context

You must be in the global configuration context: `switch(config)#`.

Parameters

`NAME`

Specifies mirror endpoint name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a mirror endpoint named test :

```
switch(config)# mirror endpoint test
```

Deleting mirror endpoint named test

```
switch(config)# no mirror endpoint test
```

show mirror

Syntax

`show mirror [<SESSION-ID>] [vsx-peer]`

Description

Shows information about mirroring sessions. If `<SESSION-ID>` is not specified, then the command shows a summary of all configured mirroring sessions. If `<SESSION-ID>` is specified, then the command shows detailed information about the specified mirroring session.

Command context

Operator (`>`) or Manager (`#`)

Parameters

`<SESSION-ID>`

Specifies the session identifier. Range: 1 to 4

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Admin Status indicates the configured status. Admin Status is one of the following values:

`enable`

The mirroring session is enabled.

`disable`

The mirroring session has been configured but not yet enabled, or has been disabled.

Operation Status indicates the status of the mirroring session. Operation Status is one of the following values:

`dest_doesnt_exist`

The configured destination interface is not found in the system. The mirroring session cannot be enabled.

`destination_shutdown`

The mirroring session is enabled, but the destination interface is shut down. No traffic can be monitored.

`disabled`

The mirroring session is disabled and is not in an error condition.

`enabled`

The mirroring session is enabled.

`external/driver_error`

An internal ASIC hardware error occurred.

`hit_active_sessions_capacity`

The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.

`internal_error`

An invalid parameter was passed to the ASIC software layer.

`no_dest_configured`

The mirroring session does not have a destination interface configured.

`no_name_configured`

A software error occurred. The mirroring session does not have a session ID in its configuration.

`null_mirror`

A software error occurred. The session object reference is invalid.

`out_of_memory`

The system is out of memory, reboot recommended.

`tunnel_route_resolution_not_populated`

If the destination tunnel IP address is not reachable.

`unknown_error`

An unexpected error occurred.

Examples

On the 6400 Switch Series, interface identification differs.

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID  Admin Status  Operation Status
--- ------------- ----------------------------------------------------
1   enable        enabled
2   disable       disabled
3   disable       disabled
4   enable        internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
 Mirror Session: 3
 Admin Status: disable
 Operation Status: disabled
 Comment: Monitor router port ingress-only traffic
 Source: interface 1/1/2 rx
 Destination: interface 1/1/3
 Output Packets: 0
 Output Bytes: 0
switch#
```

show mirror endpoint

Syntax

`show mirror endpoint [NAME]`

Description

Shows a list of all configured mirror endpoints, their Admin Status and their Operation Status.

The optional parameter will display the details of the specified mirror endpoint if it exists.

Command context

Operator (`>`) or Manager (`#`)

Parameters

`NAME`

Specifies name of the mirror endpoint instance to be displayed.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing a summary of all configured mirror endpoints on the switch:

```
switch# show mirror endpoint
Name    Admin Status   Operation Status
-----  -------------- ---------------------------------------------------
test    enable         enabled
monitor disable        disabled
```

Showing the details of enabled mirror endpoint audit:

```
switch# show mirror endpoint audit
Mirror Endpoint: audit
Admin Status: enable
Operation Status: enabled
Comment: Mirror Endpoint Audit
Type: gre
Tunnel: source 1.1.1.1 destination 1.1.1.2 id 1 vrf default
Interface: 1/1/1-1/1/10,lag1
Output Packets: 123456789
Output Bytes: 8912345678
```

shutdown

Syntax

`no shutdown`

Description

Enables mirror endpoint from its default disabled state. To verify the mirror endpoint was successfully activated, run the `show mirror endpoint NAME` command and verify that the **`Admin Status`** and **`Operational Status`** has changed from disabled to enabled. If the status value remains disabled, consult the system logs to determine the reason for activation failure. To disable the mirror endpoint, first disable the remote mirror session on the switch that's originating the data. Next, use the `shutdown` command to disable the mirror endpoint.

Command context

You must be in the global configuration context: `switch(config)#`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling a mirror endpoint:

```
switch(config)# mirror endpoint test
switch(config-mirror-endpoint-test)# no shutdown
```

Disabling a mirror endpoint:

```
switch(config)# mirror endpoint test
switch(config-mirror-endpoint-test)# shutdown
```

source

Syntax

```
source <source-id> destination <destination-id> id <1-4294967295> [vrf <VRF_NAME>] [type
{gre}]
no source
```

Description

Configures tunnel parameters of the mirror endpoint. Configuring a tunnel parameter to a mirror endpoint will replace the existing configuration. By default the VRF is `default`, users can also explicitly provide a custom VRF. The default tunnel type is considered to be GRE and users also have the option to explicitly give type as GRE.

The `no` form removes the tunnel parameters of the mirror endpoint.

Command context

You must be in the global configuration context: `switch(config)#`.

Parameters

`source-ip`

Specifies L3 encapsulated IPv4 source in the form A.B.C.D.

`destination-ip`

Specifies L3 encapsulated IPv4 destination in the form A.B.C.D.

`id`

Specifies tunnel identifier from the encapsulated packet.

`VRF_NAME`

Specifies the name of VRF for which the tunnel belongs to.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring a tunnel parameter to a mirror endpoint:

```
switch(config-mirror-endpoint-test)# source 1.1.1.1 destination 7.7.7.7 id 1 vrf
default type gre
```

source interface

Syntax

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored.

The `no` form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

Command context

`config-mirror-<SESSION-ID>`

Parameters

<PORT-NUM>

Specifies a physical port on the switch. Use the format `member/slot/port` (for example, `1/3/1`).

<LAG-NAME>

Specifies the identifier for the LAG (link aggregation group).

<DIRECTION>

Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following:

`both`

Mirror both transmitted and received packets.

`rx`

Mirror only received packets.

`tx`

Mirror only transmitted packets.

Authority

Administrators or local user group members with execution rights for this command.

Usage

There is a limit of four source interfaces in each direction of a given mirror session. However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.

When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1)# source interface
  LAG-NAME      Enter a LAG name. For example, lag10
  PORT-NUM      Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config)# mirror session 1
switch(config-mirror-1)# source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config)# mirror session 2
switch(config-mirror-2)# source interface 1/1/3 tx
switch(config-mirror-2)# source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2)# no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3)# source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1)# source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4)# source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4)# no source interface lag1 rx
```

source vlan

Syntax

```
source vlan <VLAN-NUM> {rx | tx | both}
no source vlan <VLAN-NUM> {rx | tx | both}
```

Description

Mirroring with VLAN as a source is supported in the following traffic directions:

- `both` - traffic received and transmitted
- `rx` - only received traffic
- `tx` - only transmitted traffic

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 1024 source VLANs for a given mirror session. There is also a limit of 4096 source VLANs across all mirror sessions.

Same VLAN can be configured as a mirror source for multiple sessions.

When changing a source VLAN in an enabled mirror session (i.e. adding, changing direction, or removing) mirrored packets being transmitted out of the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways.

- Reenter the `source vlan <VLAN-NUM> <direction>` command with the new preferred direction.
- Use the `no source vlan <VLAN-NUM> <direction>` form of the command with a direction (`rx` or `tx`) to selectively remove the specified direction.

Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

Mirroring allows configuration of VLAN as a source. When VLAN source is configured in the `rx` direction, all packets are mirrored as they are received in the switch. When VLAN source is configured in `tx` direction, all packets are mirrored as they are transmitted out of the switch.

For packets bridged through the switch:

- If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in `rx` direction, packets are mirrored in the pre-routed form with the Destination MAC address as the switch address.
- If the mirror is configured in `tx` direction, packets are mirrored in post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in `both` direction, one copy of the packet will be mirrored.

Control plane packets generated by the switch's CPU are processed both in theingress and the egress packet processing pipeline. The following are the behavior for mirroring with VLAN as source:

- If the mirror is configured in the `rx` or `tx` direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the `both` direction, two copies of the packets are mirrored to the mirror destination.

The `no` form command will cease mirroring traffic from the specified source VLAN and remove the source from the mirror configuration.

Command context

You must be in the global configuration context: `switch(config)#`.

Parameters

`VLAN-NUM`

Selects the VLAN number.

`direction`

Specifies the direction of mirroring. `tx` (transmit), `rx` (receive), or `both`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a mirror session and adding a VLAN as a source of traffic in both directions on that port:

```
switch# configure terminal
switch(config)# mirror session 1
switch(config-mirror-1)# source vlan 10 both
```

Creating a mirror session and adding two VLANs as sources of traffic:

directions:

```
switch# configure terminal
switch(config)# mirror session 2
switch(config-mirror-2)# source vlan 10 tx
switch(config-mirror-2)# source vlan 20 both
```

Configuring the source in session 2 to receive by specifying the source interface configuration:

```
switch(config-mirror-2)# source vlan 10 rx
```

Removing the first source interface in session 2 entirely, and removing the transmit direction from the other so that mirroring only occurs in the receive direction:

```
switch(config-mirror-2)# source vlan 10 rx
switch(config-mirror-2)# source vlan 20 tx
```

Showing maximum of 1024 mirror source VLANs allowed:

```
switch(config-mirror-2)# source vlan 2000 rx
The maximum number of source VLANs per mirror session is 1024 in each direction
```

Chapter 10
Monitoring a device using SNMP

Configuring SNMP: Refer to the *ArubaOS-CX SNMP/MIB Guide* for information on how to add SNMP so a device can be monitored from a network management system (NMS).

Configuring an SNMP trap receiver: Refer to the *ArubaOS-CX SNMP/MIB Guide* and specific information about the `show snmp trap` command to enable SNMP traps.

Chapter 11
Power-over-Ethernet

- The Power-over-Ethernet (PoE) subsystem manages power supplied to devices using standard Ethernet data cables. A Power Sourcing Equipment (PSE) supplies DC power as well as Ethernet connectivity to a Powered Device (PD) using a standard Ethernet cable. The maximum current depends on the PD Requested Class.
- A PoE subsystem contains two parts : a PSE and PD. A Power Sourcing Equipment (PSE) is a device that provides power through a standard Ethernet cable. A PoE capable switch functions as PSE. All Aruba PoE switches are considered as PSEs. A PD is a device powered by a PSE. Examples of PD are VoIP phones, Wireless APs, and IP cameras.
- When a PD or any network cable is connected to a PSE port, the PSE applies a detection voltage and measures the resistance value of the PD. If resistance is within IEEE 802.3 standard values (23 - 26k ohm), the connected device is treated as PD and classification begins. For legacy devices to be detected, you must enable prestandard detection on the switch.
- PDs are divided into different types and classes based on PD power requirements. The power supplied by the PSE is higher than the power PD draws to accommodate for the line losses that can result with the use of the standard maximum length cable(100m).
  - Type 1: PSE can supply minimum of 15.4W, and PD can draw a maximum of 13W.
  - Type 2: PSE can supply minimum of 30W, and PD can draw a maximum of 25.5W.
  - Type 3: PSE can supply minimum of 60W, and PD can draw a maximum of 51W.
  - Type 4: PSE can supply minimum of 90W, and PD can draw a maximum of 71W.
- Classes of PD:
  - Class 0: Type1 PD, it can draw a maximum of 13W.
  - Class 1: Type1 PD, it can draw a maximum of 3.84W.
  - Class 2: Type1 PD, it can draw a maximum of 6.49W.
  - Class 3: Type1 PD, it can draw a maximum of 13W.
  - Class 4: Type2 PD, it can draw a maximum of 25.5W.
  - Class 5: Type3 PD, it can draw a maximum of 40W.
  - Class 6: Type3 PD, it can draw a maximum of 51W.
  - Class 7: Type4 PD, it can draw a maximum of 62W.
  - Class 8: Type4 PD, it can draw a maximum of 71.3W.
- IEEE 802.3bt introduced 4-Pair PoE as a means of supplying higher power to PDs that need more than the current 25.5W supplied by IEEE 802.3at. To increase the available power without damaging the Ethernet cable, the standard introduced the ability to use all four pairs within the Ethernet cable instead of the two pairs used by previous standards (802.3at, 802.3af).
- Supported protocols:
  - Compatibility with IEEE 802.3af, 802.3at, 802.3bt and prestandard.
  - Long first class event supported on Type 3-4 PSE.
  - Support for Single Signature (SS) Type 0-6 and Dual Signature (DS) Type 0-4 PDs.
  - Multi-Event classification permits mutual ID of SS Class 0-6 and DS Class 0-4.

- Support LLDP Data Link Layer (DLL) Type 1-2 extension 12-octet TLV and Type 3-4 extension 29-octet TLV.
- Default PSE assigned class delivers the maximum PSE capable power at initial power up based on PD requested class.

- Always-on PoE is a feature that provides the ability for a switch to continue to provide power across user initiated reboots through software. Always-on PoE is enabled by default and no additional configuration is needed.

PDs only remain powered, no data transfer or PoE power negotiation can occur until the switch has completely booted up and in normal operation. PD faults occurring prior to full switch boot up will result in PoE power removal and restart the detection process only after switch returns to normal operation.

PoE commands

All PoE configuration commands except `threshold configuration` and `always-on poe configuration` are entered at the `config-if` context. The PoE threshold command is used at the system level whereas the `always-on poe` command is set at the slot level. These commands can only be configured in the global configuration context.

lldp dot3 poe

Syntax

```
lldp dot3 poe
no lldp dot3 poe
```

Description

Enables 802.3 TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. LLDP dot3 TLV is by default enabled for PoE.

The `no` form of this command disables 802.3 TLV list in LLDP.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

On the 6400 Switch Series, interface identification differs.

Enabling 802.3 TLV list in LLDP:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 poe
```

Disabling 802.3 TLV list in LLDP:

```
switch(config-if)# no lldp dot3 poe
```

lldp med poe

Syntax

```
lldp med poe [priority-override]
no lldp med poe [priority-override]
```

Description

Enables MED TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. Also enables the lldp-MED TLV priority to override user configured port priority for Power over Ethernet. When both dot3 and MED are enabled, dot 3 will take precedence. MED TLV is by default enabled for PoE. Priority over-ride is by default disabled.

The `no` form of this command disables MED TLV list in LLDP.

Command context

`config-if`

Parameters

`[priority-override]`

System defined name of the interface.

Authority

Administrators or local user group members with execution rights for this command.

Examples

On the 6400 Switch Series, interface identification differs.

Enabling and disabling LLDP MED PoE:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp med poe
switch(config-if)# no lldp med poe
```

Enabling and disabling LLDP MED PoE priority override:

```
switch(config-if)# lldp med poe priority-override
```

power-over-ethernet

Syntax

```
power-over-ethernet
no power-over-ethernet
```

Description

Enables per-interface power distribution. Per-port power is enabled by default with priority low. PoE cannot be disabled for individual ports when Quick PoE is enabled for the entire switch or line module.

The `no` form of this command disables per-interface power distribution.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

On the 6400 Switch Series, interface identification differs.

Enabling per-interface power distribution:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet
```

Disabling per-interface power distribution:

```
switch(config-if)# no power-over-ethernet
```

Showing Quick PoE enabled:

```
switch(config-if)# power-over-ethernet quick-poe 1/1
switch(config-if)# interface 1/1/1
switch(config-if)# no power-over-ethernet
Interface PoE cannot be disabled when Quick PoE is enabled.
```

power-over-ethernet allocate-by

Syntax

```
power-over-ethernet allocate-by {usage | class}
no power-over-ethernet allocate-by {usage | class}
```

Description

Configures the power allocation method. Power allocation method is initially based on usage. PSE Allocated power value will change to LLDP negotiated power if and when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocated Power Value will be the actual instantaneous power draw and reserve power based on actual consumption. In allocate-by class, power allocation is based on PD requested class and PSE allocated power value will be the LLDP negotiated power when LLDP exchange takes place between PSE and PD. . When there is no LLDP negotiation, PSE Allocate Power will be based on PD class. Reserve power is based on PD Class. By default, power allocation is by usage.

The `no` form of this command resets the action to default.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

On the 6400 Switch Series, interface identification differs.

Configuring the power allocation method:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet allocate-by usage
switch(config-if)# power-over-ethernet allocate-by class
```

Resetting power allocation method:

```
switch(config-if)# no power-over-ethernet allocate-by class
```

power-over-ethernet always-on

Syntax

```
power-over-ethernet always-on <MODULE-ID>
no power-over-ethernet always-on <MODULE-ID>
```

Description

Always-on PoE is a feature that provides the ability to the switch to continue to provide power across a soft reboot. It is applicable only to the interfaces which were connected and delivering before the soft reboot. Also, power will not be delivered if power to the switch is interrupted. This command enables or disables the always-on PoE feature at the switch or the slot level. By default, always-on PoE is enabled at the switch or the slot level.

The `no` form of this command disables power distribution on soft reboot.

Command context

`config`

Parameters

<MODULE-ID>

Module number to apply always-on PoE configuration.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling per-interface power distribution:

```
switch(config)# power-over-ethernet always-on 1/1
```

Disabling per-interface power distribution:

```
switch(config)# no power-over-ethernet always-on 1/1
```

power-over-ethernet assigned-class

Syntax

```
power-over-ethernet assigned-class {3 | 4 | 6}
no power-over-ethernet assigned-class
```

Description

Limit PoE power based on the assigned class. When an user assigns a maximum class to an interface, the PSE will limit the maximum power delivered to the PD up to a total power draw not exceeding the PSE assigned-class power. Power demotion occurs when a PD requested class is higher than the PSE assigned class, permitting the PD to receive power and operate in a reduced power mode. PoE ports cannot set an assigned class when Quick PoE is enabled on the sybsystem. The default assigned class is 4 for 2-pair capable PSE and 6 for 4-pair capable PSE.

The `no` form of this command resets the action to default.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

On the 6400 Switch Series, interface identification differs.

Setting PoE assigned class:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet assigned-class 4
```

Resetting PoE assigned class to default:

```
switch(config-if)# no power-over-ethernet assigned-class 4
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# interface 1/1/1
switch(config)# power-over-ethernet assigned-class 4
Interface assigned class cannot be configured when Quick PoE is enabled.
```

power-over-ethernet pre-std-detect

Syntax

```
power-over-ethernet pre-std-detect
no power-over-ethernet pre-std-detect
```

Description

Before IEEE 802.3 released the first Power over Ethernet standard (802.3af), vendors had shipped PoE capable switches and PD's. As we are backward compatible Aruba will support both IEEE standard and pre-standard 802.3af Power over Ethernet PD's concurrently. This CLI allows the user to enable or disable pre-802.3af-standard device detection and powering on the specific port. When pre-std-detect is enabled, power will be delivered on PairA only. Default is disabled.

The `no` form of this command resets the action to default.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

On the 6400 Switch Series, interface identification differs.

Enabling standard device detection:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet pre-std-detect
```

Disabling standard device detection:

```
switch(config-if)# no power-over-ethernet pre-std-detect
```

power-over-ethernet priority

Syntax

```
power-over-ethernet priority {critical | high | low}
no power-over-ethernet priority {critical | high | low}
```

Description

Sets PoE priority for an interface Specifying critical, high, or low indicates the priority of the interface in the event of power over-subscription. Within the same priority level, higher power-priority line-module ports have higher precedence. With same PoE priority and same line-module priority, lower numbered line-module ports have higher precedence. Per-interface PoE priority is low by default.

The `no` form of this command resets the priority to default PoE priority "low".

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring PoE priority:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet priority critical
switch(config-if)# power-over-ethernet priority high
```

Resetting the PoE priority to default:

```
switch(config-if)# no power-over-ethernet priority high
```

power-over-ethernet quick-poe

Syntax

```
power-over-ethernet quick-poe <MODULE-ID>
no power-over-ethernet
```

Description

Quick PoE is a feature that provides the ability for the switch to provide power to the connected powered device as soon as switch goes through cold reboot. When quick PoE is enabled on the subsystem PoE port disablement and PD demotion is not allowed. also quick PoE enablement is not allowed if any of the port is disabled on the subsystem. User should not over-subscribe the PoE power when quick PoE is enabled. Quick PoE saved configuration will work irrespective of the configuration change at reboot.

Enables quick PoE feature on the switch or the subsystem level. By default, quick-PoE is disabled for the subsystem.

The `no` form of this command disables quick PoE.

Command context

`config-if`

Parameters

<MODULE-ID>

Specifies module number for quick PoE configuration .

Authority

Administrators or local user group members with execution rights for this command.

Examples

On the 6400 Switch Series, interface identification differs.

Enabling and disabling quick PoE:

```
switch(config)# power-over-ethernet quick-poe 1/2
switch(config)# no power-over-ethernet quick-poe 1/1
```

```
switch(config-if)# power-over-ethernet quick-poe 1/1
PoE must be enabled on all interfaces before enabling Quick PoE
```

```
switch(config-if)# power-over-ethernet quick-poe 1/3
All interfaces must use the default assigned class before enabling Quick PoE
```

power-over-ethernet threshold

Syntax

```
power-over-ethernet threshold <PERCENTAGE>
no power-over-ethernet threshold <PERCENTAGE>
```

Description

Sets the threshold at which the system will send an excess power consumption notification trap. Default value is 80 percentage.

The `no` form of this command resets the action to default.

Command context

config

Parameters

<PERCENTAGE>

Excess power consumption trap threshold. Range 1-99.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the power-over-ethernet threshold:

```
switch(config)# power-over-ethernet threshold 75
```

Resetting the power-over-ethernet threshold to default:

```
switch(config-if)# no power-over-ethernet threshold 75
```

power-over-ethernet priority

Syntax

```
power-over-ethernet trap
no power-over-ethernet trap
```

Description

This command enables/disables the SNMP trap generation for PoE related events at system level. PoE trap generation is enabled by default.

The `no` form of this command resets the priority to default PoE priority "low".

Command context

config-if

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling SNMP trap generation for PoE:

```
switch(config)# power-over-ethernet trap
```

Disabling SNMP trap generation for PoE:

```
switch(config-if)# no power-over-ethernet trap
```

show lldp local

Syntax

```
show lldp local-device [<INTERFACE-ID>]
```

Description

Displays information advertised by the switch if the LLDP feature is enabled by user.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface. Format: `member/slot/port`

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

On the 6400 Switch Series, interface identification differs.

Showing LLDP local device:

```
switch# show lldp local-device 1/1/10
Local Port Data
===============

Port-ID            : 1/1/10
Port-Desc          : "1/1/10"
Port VLAN ID       : 0

PoE Plus Information

PoE Device Type     : Type 2 PSE
Power Source        : Primary
Power Priority      : low
PSE Allocated Power: 25.0 W
PD Requested Power : 25.0 W
```

show lldp neighbor

Syntax

```
show lldp neighbor [<INTERFACE-ID>]
```

Description

Displays detailed information about a particular neighbor connected to a particular interface.

Command context

Operator (>) or Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface. Format: `member/slot/port`

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

On the 6400 Switch Series, interface identification differs.

Showing LLDP neighbor information when there is only one neighbor:

```
switch# show lldp neighbor-info 1/1/10

Port                           : 1/1/10
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description   : ArubaOS (MODEL: 325), Version Aruba IAP
Neighbor Chassis-ID            : 84:d4:7e:ce:5d:68
Neighbor Management-Address    : 169.254.41.250
Chassis Capabilities Available : Bridge, WLAN
Chassis Capabilities Enabled   :
Neighbor Port-ID               : 84:d4:7e:ce:5d:68
Neighbor Port-Desc             : eth0
TTL                            : 120
Neighbor Port VLAN ID          :
Neighbor PoEplus information   : DOT3
Neighbor Device Type           : TYPE2 PD
Neighbor Power Priority        : Unkown
Neighbor Power Source          : Primary
Neighbor Power Requested       : 25.0 W
Neighbor Power Allocated       : 0.0 W
Neighbor Power Supported       : No
Neighbor Power Enabled         : No
Neighbor Power Class           : 5
Neighbor Power Paircontrol     : No
Neighbor Power Pairs           : SIGNAL
```

show power-over-ethernet

Syntax

6300 Switch Series:

`show power-over-ethernet [member <MEMBER-ID>] [brief]`

6400 Switch Series:

`show power-over-ethernet [<MODULE-ID>] [brief]`

6300, 6400 Switch Series:

`show power-over-ethernet [<IFRANGE> [brief]`

Description

Displays the status information of the full system. Displays the brief status of all port or given port if parameter brief is used. Displays the detailed status of given port.

Command context

Operator (>) or Manager (#)

Parameters

<MEMBER-ID>

Displays the detailed status of given member.

<MODULE-ID>

Displays detailed status for the given module.

<IFRANGE>

Port identifier range.

<IFNAME>

Display the detailed status of given port.

`brief`

Display the brief status of all ports or the given port.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing sample output for show power-over-ethernet on standalone box with VSF capabiity:

```
switch# show power-over-ethernet

System Power Status for member 1

  Configured  Power Status     : No redundancy
  Operational Power Status     : No redundancy
  Total Available Power        : 740 W
  Total Failover Pwr Avl       :   0 W
  Total Redundancy Power       :   0 W
  Total Power Drawn            :   0 W +/- 6W
  Total Power Reserved         :   0 W
  Total Remaining Power        : 740 W
  Trap Threshold               : 80 %
  Trap Enabled                 : Yes
  Always-on PoE Enabled        : 1/1
  Quick PoE Enabled            : None


Internal Power
        Total Power
  PS    (Watts)        Status
  ----- -------------  ---------------------
  1     0              Absent
  2     740            Ok

System Power Status for member 2

  Configured  Power Status     : No redundancy
  Operational Power Status     : No redundancy
  Total Available Power        : 600 W
  Total Failover Pwr Avl       :   0 W
  Total Redundancy Power       :   0 W
  Total Power Drawn            :   0 W +/- 6W
  Total Power Reserved         :   0 W
  Total Remaining Power        : 600 W
  Trap Threshold               : 80 %
  Trap Enabled                 : Yes
  Always-on PoE Enabled        : None
  Quick PoE Enabled            : None
```

```
Internal Power
        Total Power
  PS     (Watts)        Status
  ----- -------------  ---------------------
  1     0              Absent
  2     600            Ok
```

Showing sample output for power-over-ethernet member:

```
switch# show power-over-ethernet member 1

System Power Status for member 1

  Configured  Power Status      : No redundancy
  Operational Power Status      : No redundancy
  Total Available Power         : 740 W
  Total Failover Pwr Avl        :   0 W
  Total Redundancy Power        :   0 W
  Total Power Drawn             :   0 W +/- 6W
  Total Power Reserved          :   0 W
  Total Remaining Power         : 740 W
  Trap Threshold                : 80 %
  Trap Enabled                  : No
  Always-on PoE Enabled         : 1/1
  Quick PoE Enabled             : 1/1

Internal Power
        Total Power
  PS     (Watts)        Status
  ----- -------------  ---------------------
  1     0              Absent
  2     740            Ok
```

Showing sample output for power-over-ethernet brief in a VSF stack:

```
switch# show power-over-ethernet brief

Status and Configuration Information for PoE

  Member 1 Power Status
    Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
    Always-on PoE Enabled: 1/1
    Quick PoE Enabled: None

PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port      PD     Cls Type
Port     En  Priority Detect  Act   Rsrvd   Draw   Status        Sign
-------  --- ------   ------- ----- ------  ------ ---------     -----  --- ----
1/1/1    Yes Low      Off     Class  0.0 W   0.0 W Denied        None   4   2
1/1/2    Yes Critical Off     Usage  1.6 W   1.5 W Delivering*   Single 0   1
1/1/3    Yes High     Off     Class 54.0 W  25.5 W Delivering*^  Dual   1/3 3
1/1/4    No  Low      On      Usage  0.0 W   0.0 W Disabled      None   N/A N/A

  Member 2 Power Status
    Available: 600 W  Reserved: 0.00 W  Remaining: 600 W
    Always-on PoE Enabled: None
    Quick PoE Enabled: None

PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port      PD     Cls Type
```

```
Port     En  Priority Detect  Act   Rsrvd   Draw   Status        Sign
------- --- ------   ------- ----- ------  ------ ---------     -----  --- ----
2/1/1    Yes Low      Off     Class  0.0 W   0.0 W Searching     None   N/A  N/A
2/1/2    Yes Critical Off     Usage  0.0 W   0.0 W Searching     None   N/A  N/A
2/1/3    Yes High     Off     Class  0.0 W   0.0 W Searching     None   N/A  N/A
2/1/4    No  Low      On      Usage  0.0 W   0.0 W Disabled      None   N/A  N/A

*This port may go down in the event of a PSU failure.
^This port is power demoted due to user config or power availabilty.
```

Showing sample output for power-over-ethernet brief for a Chassis system:

```
switch# show power-over-ethernet brief

Status and Configuration Information for PoE

  Power Status
    Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
    Always-on PoE Enabled: 1/1,1/3,1/4,1/7
    Quick PoE Enabled: None

PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port      PD     Cls Type
Port     En  Priority Detect  Act   Rsrvd   Draw   Status        Sign
------- --- ------   ------- ----- ------  ------ ---------     -----  --- ----
1/1/1    Yes Low      Off     Class  0.0 W   0.0 W Denied        None   4   2
1/1/2    Yes Critical Off     Usage  1.6 W   1.5 W Delivering*  Single 0   1
1/1/3    Yes High     Off     Class 54.0 W  25.5 W Delivering^  Dual   1/3 3
1/1/4    No  Low      On      Usage  0.0 W   0.0 W Disabled      None   N/A N/A

*This port may go down in the event of a PSU failure.
^This port is power demoted due to user config or power availabilty.
```

Showing sample output for power-over-ethernet brief per-port:

```
switch# show power-over-ethernet 1/1/1 brief

Status and Configuration Information for port 1/1/1

  Member 1Power Status
    Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
    Always-on PoE Enabled: 1/1
PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port      PD     Cls Type
Port     En  Priority Detect  Act   Rsrvd   Draw   Status        Sign
------- --- ------   ------- ----- ------  ------ ---------     -----  --- ----
1/1/1    Yes Low      Off     Class  0.0 W   0.0 W Denied        None   4   2
```

Showing sample output for power-over-ethernet brief for interface range:

```
switch# show power-over-ethernet 1/1/1-1/1/2 brief

Status and Configuration Information for port 1/1/1-1/1/2

  Member 1Power Status
    Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
    Always-on PoE Enabled: 1/1
PoE      Pwr Power    Pre-std Alloc PSE Pwr PD Pwr PoE Port      PD     Cls Type
Port     En  Priority Detect  Act   Rsrvd   Draw   Status        Sign
```

```
-------  --- ------   ------- ----- ------  ------ ---------    -----  --- ----
1/1/1    Yes Low      Off     Class  0.0 W   0.0 W Denied       None   4   2
1/1/2    Yes Critical Off     Usage  1.6 W   1.5 W Delivering*  Single 0   1
```

Showing sample output for power-over-ethernet for a missing line card:

```
switch# show power-over-ethernet 1/3 brief

Module 1/3 is not physically present.
```

Showing sample output for power-over-ethernet brief for a missing member:

```
switch# show power-over-ethernet member 3 brief

Member 3 is not physically present.
```

Showing sample output for power-over-ethernet port when physical interface is not present:

```
switch# show power-over-ethernet 2/1/1

Interface 2/1/1 is not present.
```

Chapter 12
Aruba AirWave

You can manage and monitor the AOS-CX switch through Aruba AirWave. The following benefits and functions include:

- Configuration (partial configuration)
- Device topology
- Immediate and historical trend reports
- Monitoring of the device and user connected to the network.
- Network discovery
- Syslogs and trap receiver

For information about which versions of Aruba AirWave support AOS-CX, see the *ArubaOS-CX Release Notes*.

SNMP support and AirWave

For AirWave to discover and monitor the switch, you must:

- Enable the SNMP services on the switch.
- Configure the SNMP agent to use the SNMP version supported by the management station.

SNMP on the switch

The switch provides SNMP services through the management channel and the data interfaces. Functionality, such as device discovery from NMS, syslog and trap forwarding, can be any channel configured by you.

Although the SNMP server can be enabled on both VRFs (`mgmt` and `default`), only one instance of SNMP can be running. The highest priority is on the `default` VRF.

For example, assume that SNMP is first enabled on the `mgmt` VRF (`snmp-server vrf mgmt`). Then, SNMP is enabled on the `default` VRF (`snmp-server vrf default`) without disabling SNMP on the `mgmt` (using an equivalent `no` form of the command). The `show running-config` command displays both `snmp-server vrf` commands; however, the SNMP instance is running only on the `default` VRF (highest priority).

```
switch# config
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
switch(config)# show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.01.
led locator on
!
!
!
snmp-server vrf default
snmp-server vrf mgmt
!
...
```

Supported features with AirWave and the AOS-CX switch

AirWave supports the following features with the AOS-CX switch:

Device management	Device discovery using SNMPv2C and SNMPv3
	Device dashboards
Monitoring management	Device health attributes (device status/reachability)
	Interface and VLAN management
	Initiates an SSH connection from Aruba AirWave to AOS-CX so that the device outputs from the AOS-CX CLI can be displayed in the Aruba AirWave user interface.
	Firmware versions
	Displays neighbor devices connected to AOS-CX switches
	Device topology
Configuration management	Partial configuration
Alarm management	Alarm triggers (device and interface up/down, new device discoveries, custom event triggers)
	Syslogs and traps
Report management	Device inventory, interface utilization, and device reachability reports
	Summary report of device model, firmware, and boot loader version

Configuring the AOS-CX switch to be monitored by AirWave

Prerequisites

Aruba AirWave is active on the network.

Procedure

1. Enable SNMP on the ArubaOS-CX switch by entering the snmp-server vrf command.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
```

2. Configure the SNMPv2C community to public by entering the `snmp-server community public` command. In this instance, `public` is a read-only community string.

```
switch(config)# snmp-server community public
```

3. The community-string is used by SNMPv1 and SNMPv2C for unencrypted authentication. SNMPv3 lets you encrypt the authentication mechanism. To enable SNMPv3, enter the `snmpv3 user` and

snmpv3 context commands.

```
switch(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=

switch(config)# snmpv3 context Admin
```

For discovering devices in AirWave through the SNMPv3 community, the SNMPv3 context name is not mandatory. Devices can still be discovered in Aruba AirWave without the SNMPv3 context name.

4. Enter the logging command for enabling syslog forwarding to a remote syslog server, such as AirWave:

```
switch(config)# logging 10.0.10.2 severity debug
```

5. SNMP traps enable an agent to notify the management station of significant events by way of an unsolicited SNMP message. Enable SNMP traps by entering the snmp-server host command:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c vrf default
```

SNMP traps cannot be forwarded from AOS-CX 10.00 switches that have the VRF configured as mgmt. Later versions of AOS-CX support SNMP trap forwarding even when the VRF is configured as default or mgmt.

6. For information on how to add a device for monitoring in the Aruba AirWave user interface, see the documentation for Aruba AirWave.

logging

Syntax

```
logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}
     [udp [<PORT-NUM>] | tcp [<PORT-NUM>] | tls [<PORT-NUM>]]
     [include-auditable-events] [severity <LEVEL>] [vrf <VRF-NAME>]
logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}
     [tls [<PORT-NUM>]] [auth-mode {certificate | subject-name}]
     [legay-tls-renegotiation] [include-auditable-events] [severity <LEVEL>]
     [vrf <VRF-NAME>]
no logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>
```

Description

Enables syslog forwarding to a remote syslog server.

The no form of this command disables syslog forwarding to a remote syslog server.

Command context

config

Parameters

{<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}

Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.

[udp [<PORT-NUM>] | tcp [<PORT-NUM>]]

Specifies the UDP port or TCP port of the remote syslog server to receive the forwarded syslog messages.

`udp [<PORT-NUM>]`

Range: 1 to 65535. Default: 514

`tcp [<PORT-NUM>]`

Range: 1 to 65535. Default: 1470

`tls [<PORT-NUM>]`

Range: 1 to 65535. Default: 6514

`include-auditable-events`

Specifies that auditable messages are also logged to the remote syslog server.

`severity <LEVEL>`

Specifies the severity of the syslog messages:

- `alert`: Forwards syslog messages with the severity of `alert (6)` and `emergency (7)`.
- `crit`: Forwards syslog messages with the severity of `critical (5)` and above.
- `debug`: Forwards syslog messages with the severity of `debug (0)` and above.
- `emerg`: Forwards syslog messages with the severity of `emergency (7)` only.
- `err`: Forwards syslog messages with the severity of `err (4)` and above
- `info`: Forwards syslog messages with the severity of `info (1)` and above. Default.
- `notice`: Forwards syslog messages with the severity of `notice (2)` and above.
- `warning`: Forwards syslog messages with the severity of `warning (3)` and above.

`auth-mode`

Specifies the TLS authentication mode used to validate the certificate.

- certificate: Validates the peer using trust anchor certificate based authentication. Default.
- subject-name: Validates the peer using trust anchor certificates as well as subject-name based authentication.

`legacy-tls-renegotiation`

Enables the TLS connection with a remote syslog server supporting legacy renegotiation.

`vrf <VRF-NAME>`

Specifies the VRF used to connect to the syslog server. Optional. Default: `default`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of `err (4)` and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF `lab_vrf`:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)# logging example.com tls auth-mode subject name
```

snmp-server community

Syntax

```
snmp-server community <STRING>
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is a password that controls read access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The `no` form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value `public` is automatically defined.

Command context

`config`

Parameters

<STRING>

Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

snmp-server host

Syntax

```
snmp-server host <IPv4-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]

no snmp-server host <IPv4-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
```

```
snmp-server host <IPv4-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]

no snmp-server host <IPv4-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]

snmp-server host <IPv4-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]

no snmp-server host <IPv4-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The `no` form of this command removes the specified trap/inform receiver.

Configuring `snmpv3 informs` is not supported.

Command context

`config`

Parameters

<IPv4-ADDR>

Specifies the IP address of a trap receiver in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. You can remove leading zeros. For example, the address `192.169.005.100` becomes `192.168.5.100`.

`trap version` *<VERSION>*

Specifies the trap notification type for SNMPv1 or v2c. Available options are: `v1` or `v2c`.

`inform version v2c`

Specifies the inform notification type for SNMPv2c.

`trap version v3`

Specifies the trap notification type for SNMPv3.

`user` *<NAME>*

Specifies the SNMPv3 user name to be used in the SNMP trap notifications.

`community` *<STRING>*

Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: `public`.

<UDP-PORT>

Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.

`vrf` *<VRF-NAME>*

Specifies the name of the VRF on which to send the notifications.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
```

```
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public port
5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public port
5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public
port 5000 vrf default

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
```

snmp-server vrf

Syntax

```
snmp-server vrf <VRF-NAME>
no snmp-server vrf <VRF-NAME>
```

Description

Configures the VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF.

The `no` form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Command context

`config`

Parameters

<VRF-NAME>

Specifies the VRF on which the SNMP agent listens for incoming requests. The SNMP agent can listen on either the `mgmt` or `default` VRF. If configured for both, the SNMP agent listens on `default`, which has a higher priority.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# snmp-server vrf default
```

```
switch(config)# no snmp-server vrf default
```

snmpv3 context

Syntax

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]
no snmpv3 context <NAME> [vrf <VRF-NAME>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The `no` form of this command removes the specified SNMP context.

Command context

`config`

Parameters

`<NAME>`

Specifies the name of the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark (?).

`vrf <VRF-NAME>`

Specifies the VRF associated with the context. Default: default.

`community <STRING>`

Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Creating an SNMPv3 context named **newContext** on VRF **myVrf** and with community string **private**.

```
switch(config)# snmpv3 context newContext vrf myVrf community private
```

Removing the SNMPv3 context named **newContext** on VRF **myVrf**:

```
switch(config)# no snmpv3 context newContext vrf myVrf
```

snmpv3 user

Syntax

```
snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass {plaintext | ciphertext}
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass {plaintext | ciphertext} <PRIV-PWORD>] ]
```

```
no snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass <PRIV-PWORD>] ]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context.

The `no` form of this command removes the specified SNMPv3 user.

Command context

`config`

Parameters

<NAME>

Specifies the SNMPv3 username. Range 1 - 32 printable ASCII characters, excluding space and question mark.

`auth` *<AUTH-PROTOCOL>*

Specifies the authentication protocol used to validate user logins. Available options are: `md5` or `sha`.

`auth-pass {plaintext | ciphertext}` *<AUTH-PWORD>*

Specifies the SNMPv3 user password. Range for `plaintext` is 8 - 32 printable ASCII characters, excluding space and question mark.

Range for `ciphertext` is 1 - 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

`priv` *<PRIV-PROTOCOL>*

Specifies the SNMPv3 security protocol (encryption method). Available options are: `aes` or `des`.

`priv-pass {plaintext | ciphertext}` *<PRIV-PWORD>*

Specifies the SNMPv3 user privacy passphrase. Range for `plaintext` is 8 - 32 printable ASCII characters, excluding space and question mark.

Range for `ciphertext` is 1 - 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
```

Removing an SNMPv3 user named `Admin`:

```
switch(config)# no snmpv3 user Admin
```

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
```

Copying an SNMP user from switch 1 to switch 2.

On switch 1, configure a user called **Admin**, then issue the `show running-config` command to display switch configuration settings. The snmpv3 user command uses the `ciphertext` option to protect the users's passwords.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword
priv des priv-pass plaintext myprivpass
switch1(config)# exit
switch1# show running-config
Current configuration:
!
!Version ArubaOS-CX TL.10.00.0003-8017-gdeb0606~dirty
!
!
!
snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
ssh server vrf mgmt
!
!
!
!
interface mgmt
    no shutdown
    ip dhcp
vlan 1
```

On switch 2, execute the snmpv3 user command that was displayed by `show running-config` on switch 1. This creates the user on switch 2 with the same configuration settings.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass
ciphertextAQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=priv
des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
```

Chapter 13
Support and other resources

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

Aruba Support Portal

https://asp.arubanetworks.com/downloads

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

https://www.hpe.com/networking/support

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

https://support.hpe.com/portal/site/hpsc/aae/home/

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://asp.arubanetworks.com/notifications/subscriptions (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty information

To view warranty information for your product, go to https://www.arubanetworks.com/support-services/product-warranties/.

Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see https://www.arubanetworks.com/company/about-us/environmental-citizenship/.

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